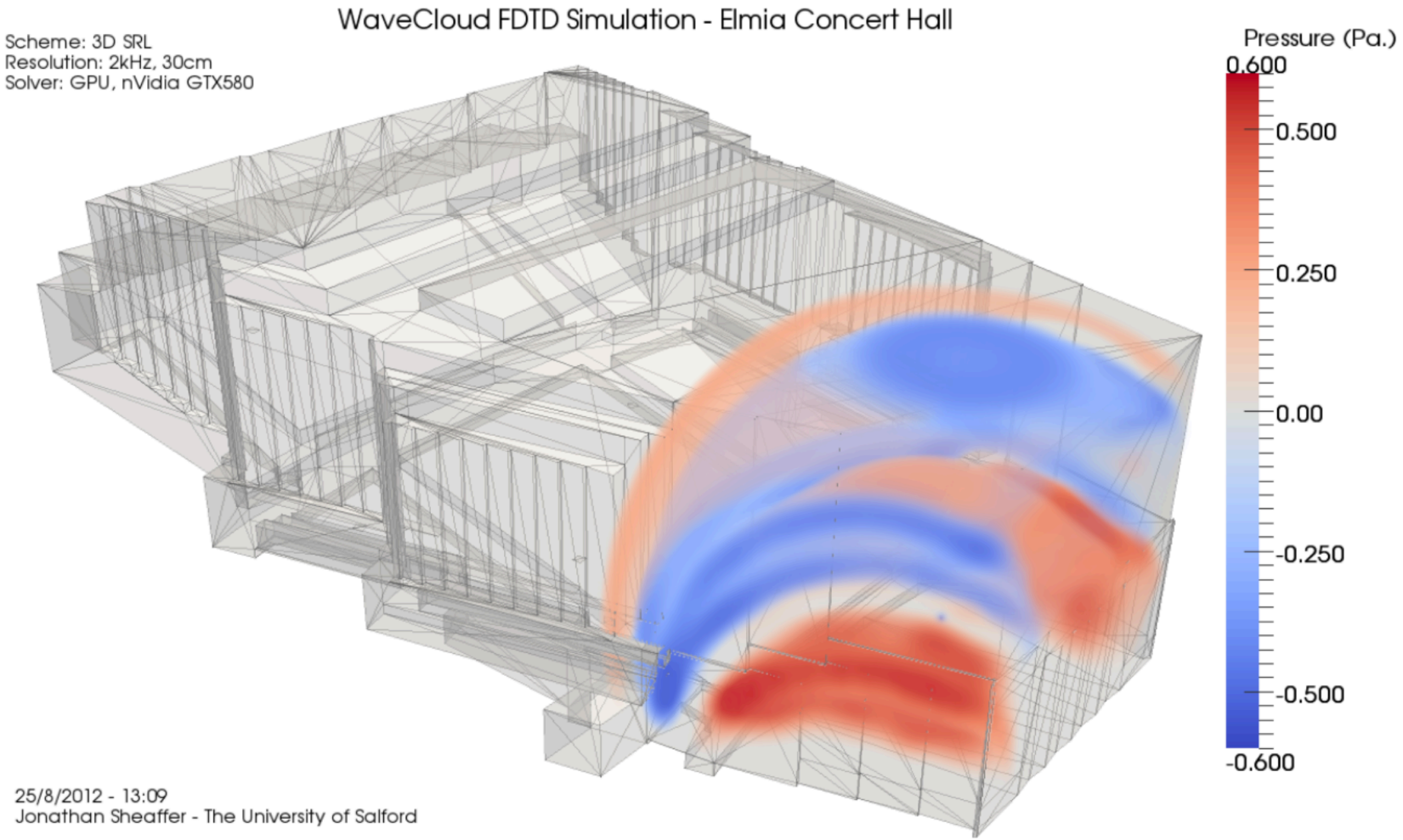
# Jet marching on unstructured meshes: algorithms and applications

Samuel F. Potter

Coreform, LLC

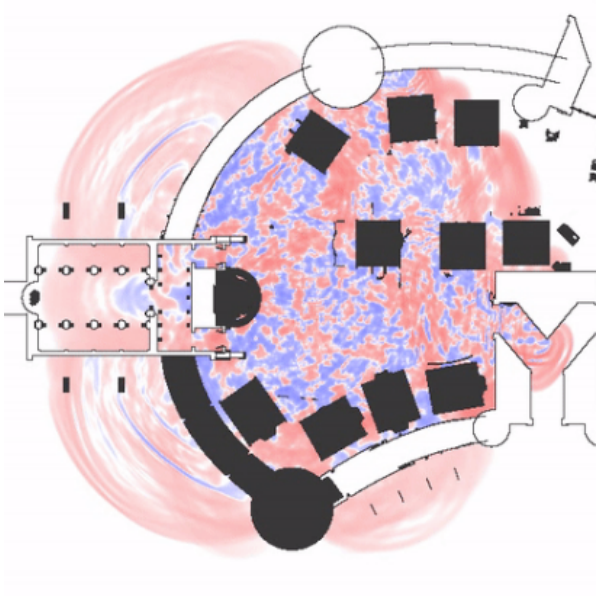
# Introduction

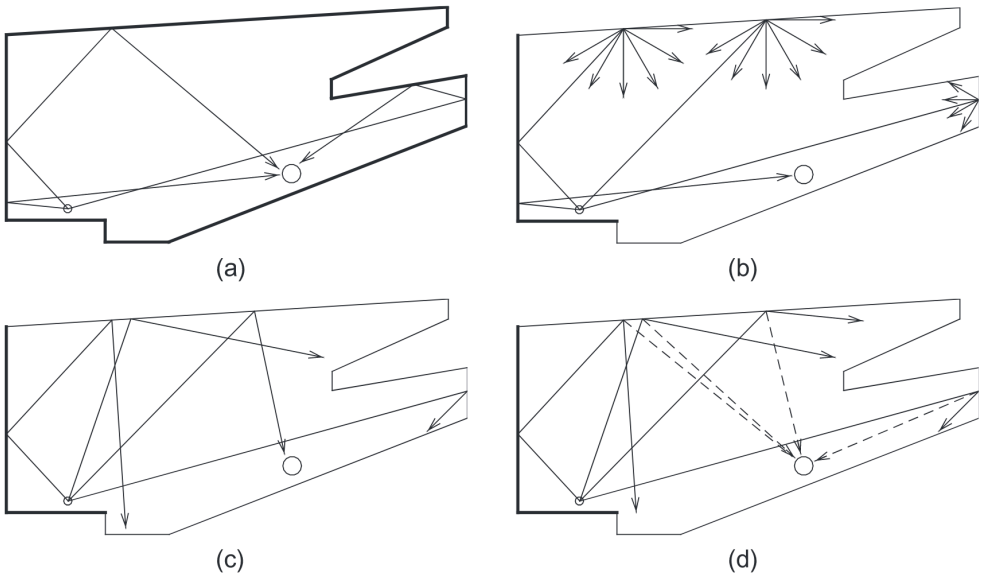
# Introduction: computational room acoustics




Sheaffer, 2013.

# Introduction: computational room acoustics






**Numerical acoustics**: solve wave equation for bandlimited impulse input. High-frequency (>2kHz) simulations expensive and impractical.

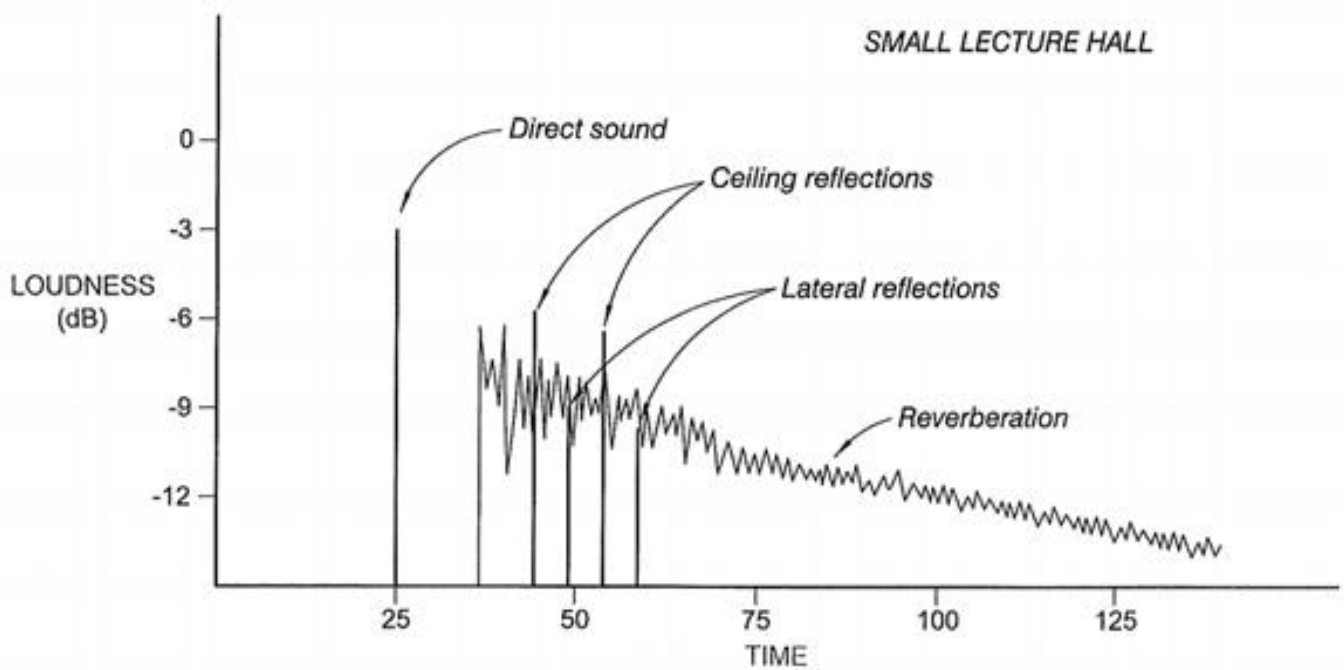
**Geometric acoustics**: use raytracing to assemble room impulse response. Handling diffraction effects with raytracing is complicated.

Left: `https://www.microsoft.com/en-us/research/project/project-triton`

Right: Kuttruff, 2016.

# Introduction: the room impulse response

**Basic task:** for a pair of points in the room (a *source* and *listener*), compute or encode the room impulse response. This is the Green's function for the wave equation for a *fixed sound source location*.




Source: https://sites.google.com/site/roomacgroup4/

# Introduction: PDEs for acoustics

Wave equation for pressure $p = p(\boldsymbol{x}, t)$:

$$p_{tt} = c^2 \Delta p$$

Time-harmonic solution for frequency $\omega$:

$$p(\boldsymbol{x}, t) = e^{i\omega t} u_\omega(\boldsymbol{x})$$

Leads to the Helmholtz equation:

$$\Delta u_\omega + k^2 u_\omega = 0, \qquad k = \frac{\omega}{c}$$

where $c$ is the speed of sound and $k$ is the wavenumber.

# Introduction: high-frequency wave propagation

Geometric optics ansatz ("WKB") for Helmholtz:

$$u_\omega(\boldsymbol{x}) \sim A_\omega(\boldsymbol{x}) e^{i\omega\tau(\boldsymbol{x})}$$

Leads to the eikonal equation:

$$\|\nabla\tau\| = 1/c = s, \qquad s \text{ is the "slowness"}$$

and the amplitude equation:

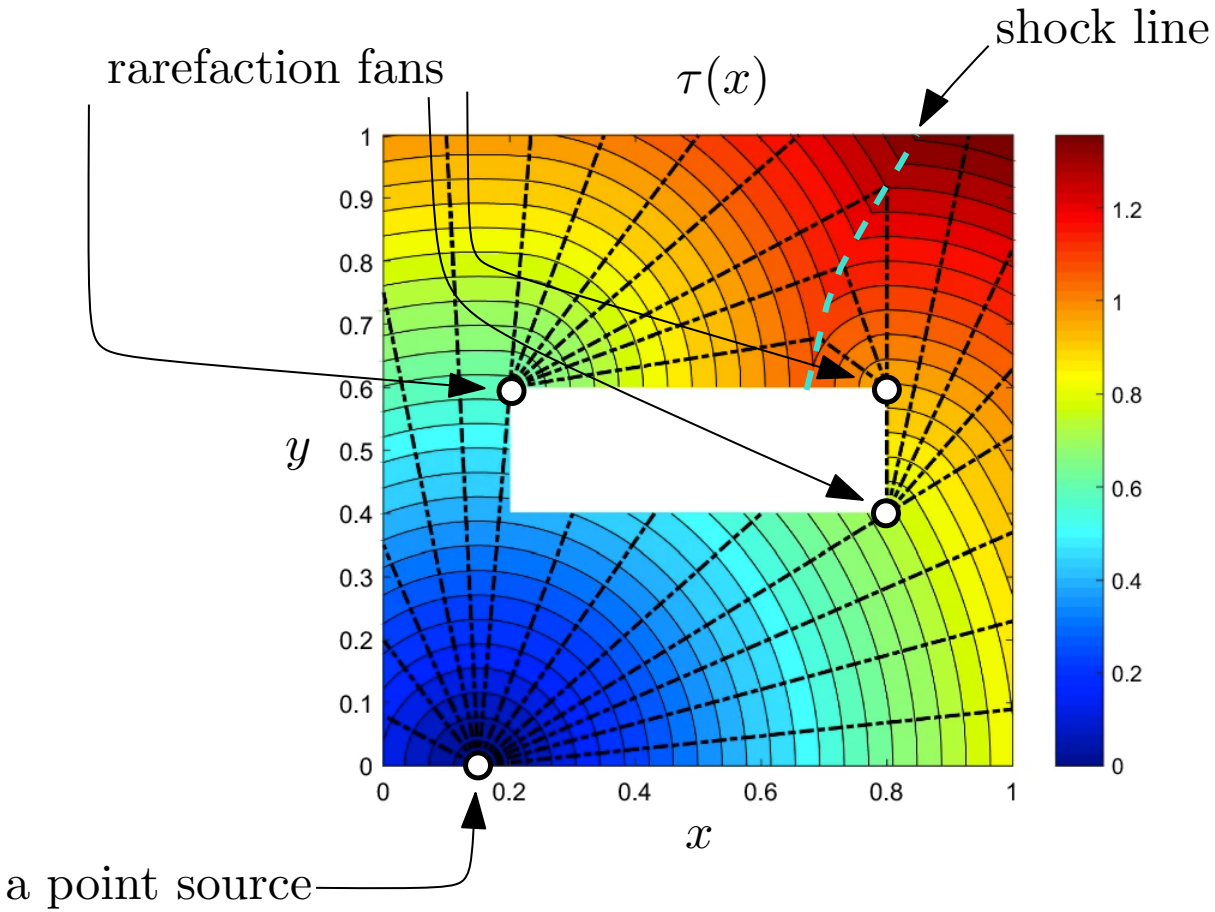
$$\nabla A_\omega \cdot \nabla\tau + 2A_\omega \Delta\tau = 0$$

High-frequency approximation of the Helmholtz equation:

$$|u_\omega - A_\omega e^{i\omega\tau}| = O(\omega^{-1})$$

# Introduction: the eikonal equation, $\|\nabla\tau\| = 1/c$

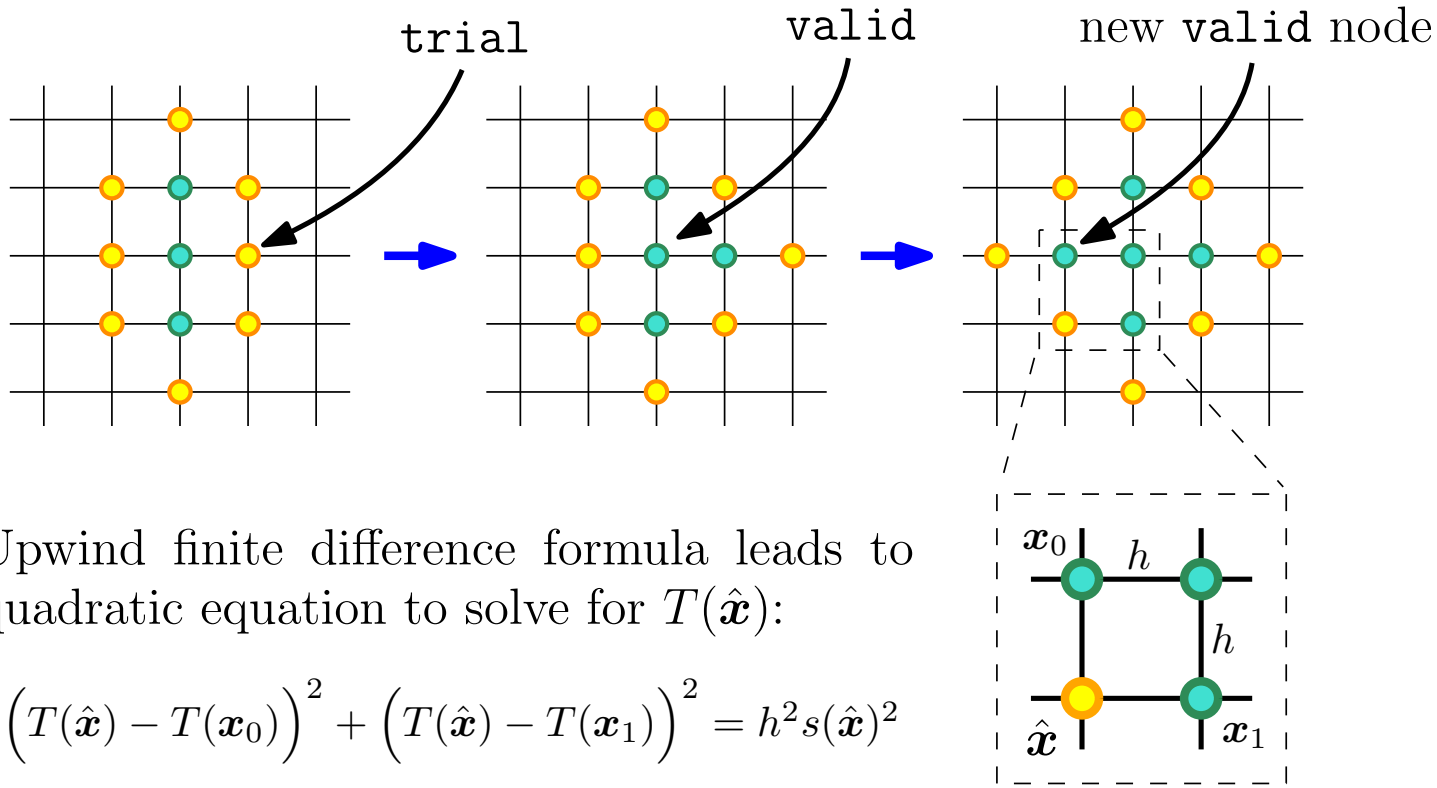
A PDE whose solution encodes only the first arrival time accounting for diffraction. Its characteristics are rays.




Source: Qi & Vladimirsky, 2019.

# Introduction: the fast marching method

A single-pass, $O(h)$ accurate, upwind finite difference method for solving the eikonal equation based on Dijkstra's algorithm




Upwind finite difference formula leads to quadratic equation to solve for $T(\hat{\boldsymbol{x}})$:

$$\Big(T(\hat{\boldsymbol{x}}) - T(\boldsymbol{x}_0)\Big)^2 + \Big(T(\hat{\boldsymbol{x}}) - T(\boldsymbol{x}_1)\Big)^2 = h^2 s(\hat{\boldsymbol{x}})^2$$

All `trial` nodes ordered in a priority queue by current $T$ value

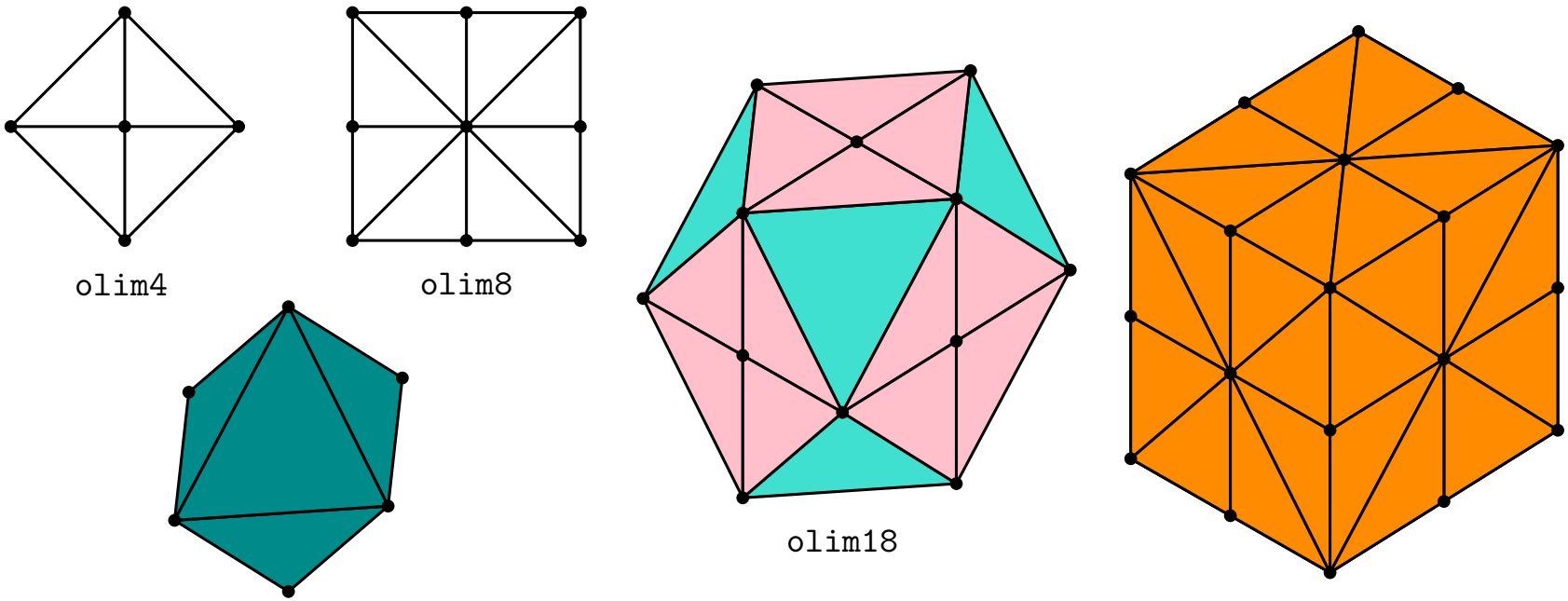
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Sethian, 1996.

# Introduction: semi-Lagrangian marchers

Can also solve the eikonal equation by combining local raytracing (semi-Lagrangian) with marching (Dijkstra). Ex: Tsitsiklis's algorithm, **OLIMs**...

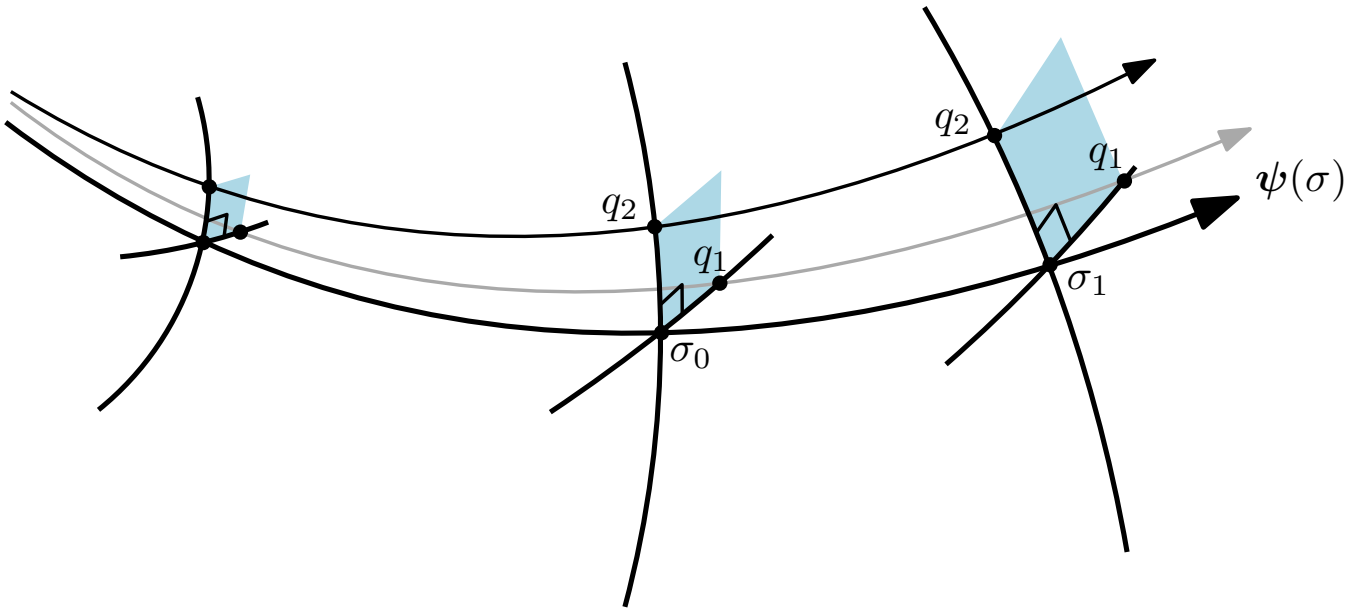
$$T(\hat{\boldsymbol{x}}) = \min_{\boldsymbol{\psi}} \left\{ T(\boldsymbol{\psi}_0) + \int_{\boldsymbol{\psi}} s d\ell \right\}$$




Potter and Cameron, 2019.

# Introduction: the amplitude equation

Using **paraxial raytracing**, the amplitude can be computed in terms of the local geometric spreading of a ray tube:




The amplitude along a ray $\psi$ satisfies:

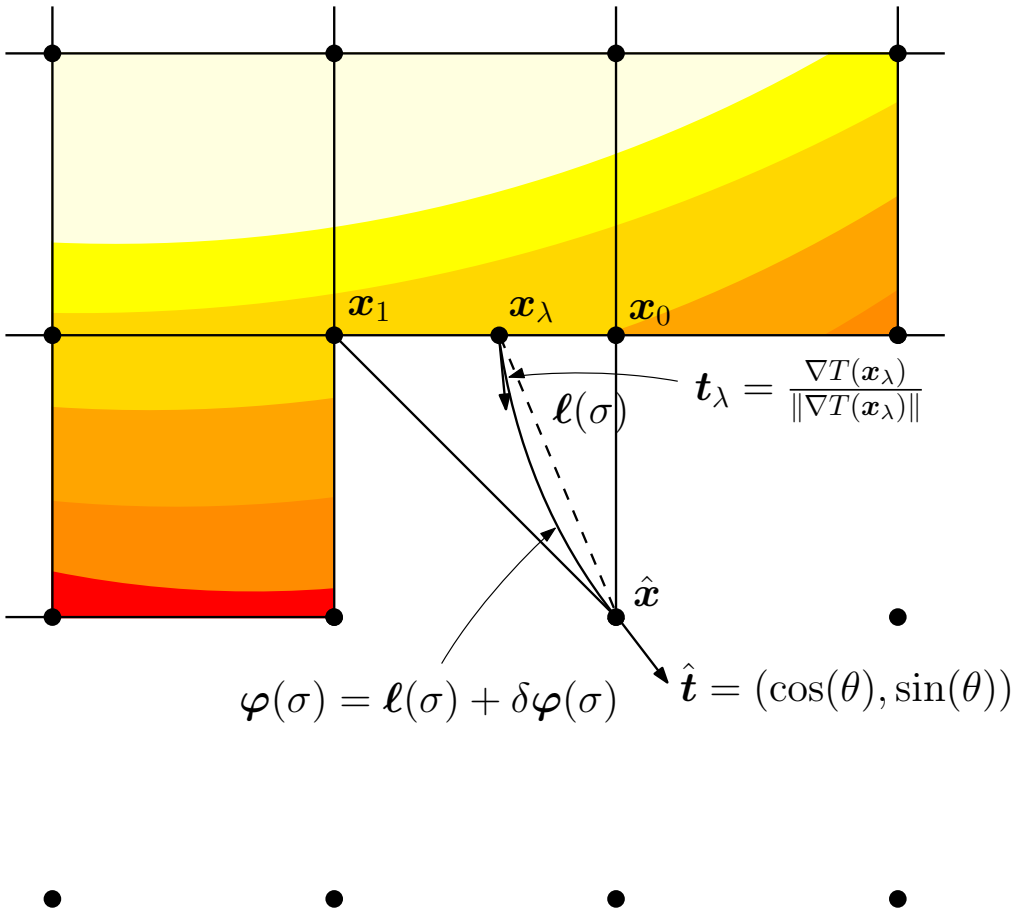
$$\alpha(\psi(\sigma_1)) = \alpha(\psi(\sigma_0))\sqrt{\frac{s(\psi(\sigma_1))}{|\det(Q(\sigma_1))|}}, \qquad Q = \begin{bmatrix} q_1 & q_2 \end{bmatrix}$$

Popov, 1978.

# Jet marching methods (JMMs) in 2D

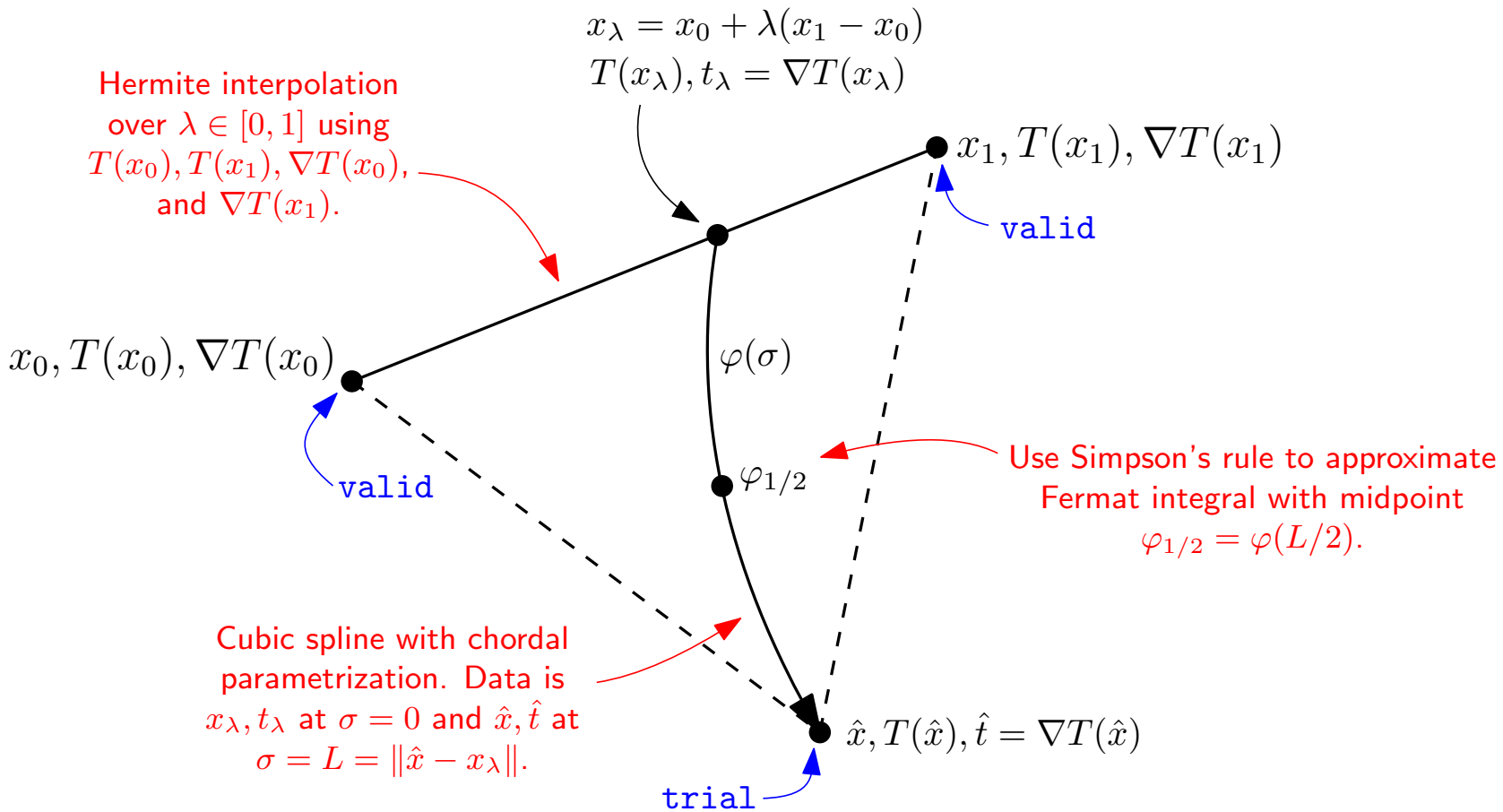
# 2D JMMs: overview

Approximate $T$ and $\psi$ using cubic polynomials, march the **jet** of the eikonal (coefficients of truncated Taylor polynomial):




Potter and Cameron, 2021.

# 2D JMMs: higher-order updates in 2D

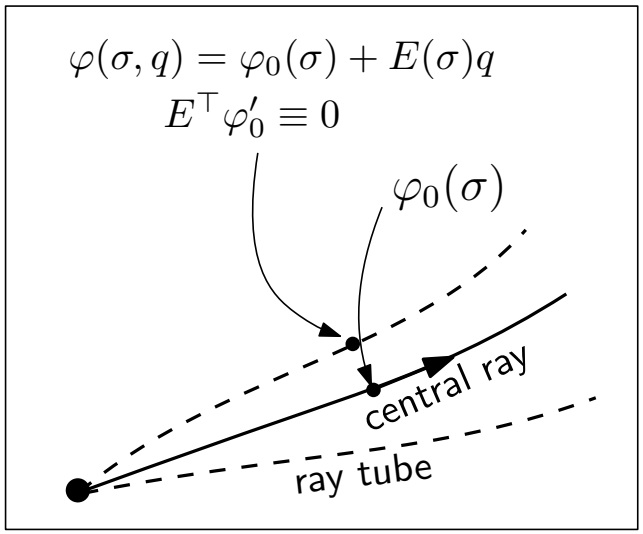



$$T(\hat{x}) = \min_{\substack{\hat{t}\in\mathbb{S}^1 \\ 0\le\lambda\le 1}} \left\{ \mathsf{T}(\lambda) + \frac{L}{6}\left( s(\hat{x}) + 4s(\varphi(\tfrac{L}{2}))\|\varphi'(\tfrac{L}{2})\| + s(x_\lambda) \right) \right\}$$

Potter and Cameron, 2021.

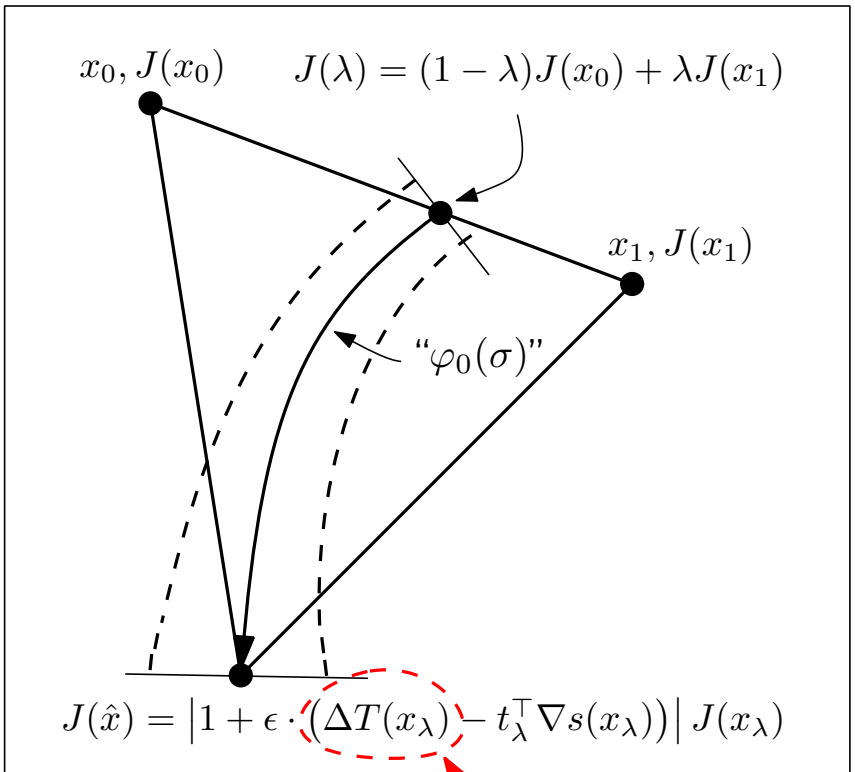
# 2D JMMs: marching the geometric spreading

Linearly approximate the geometric spreading $J$ over $[\boldsymbol{x}_0, \boldsymbol{x}_1]$ and march it after parametrizing the local ray:



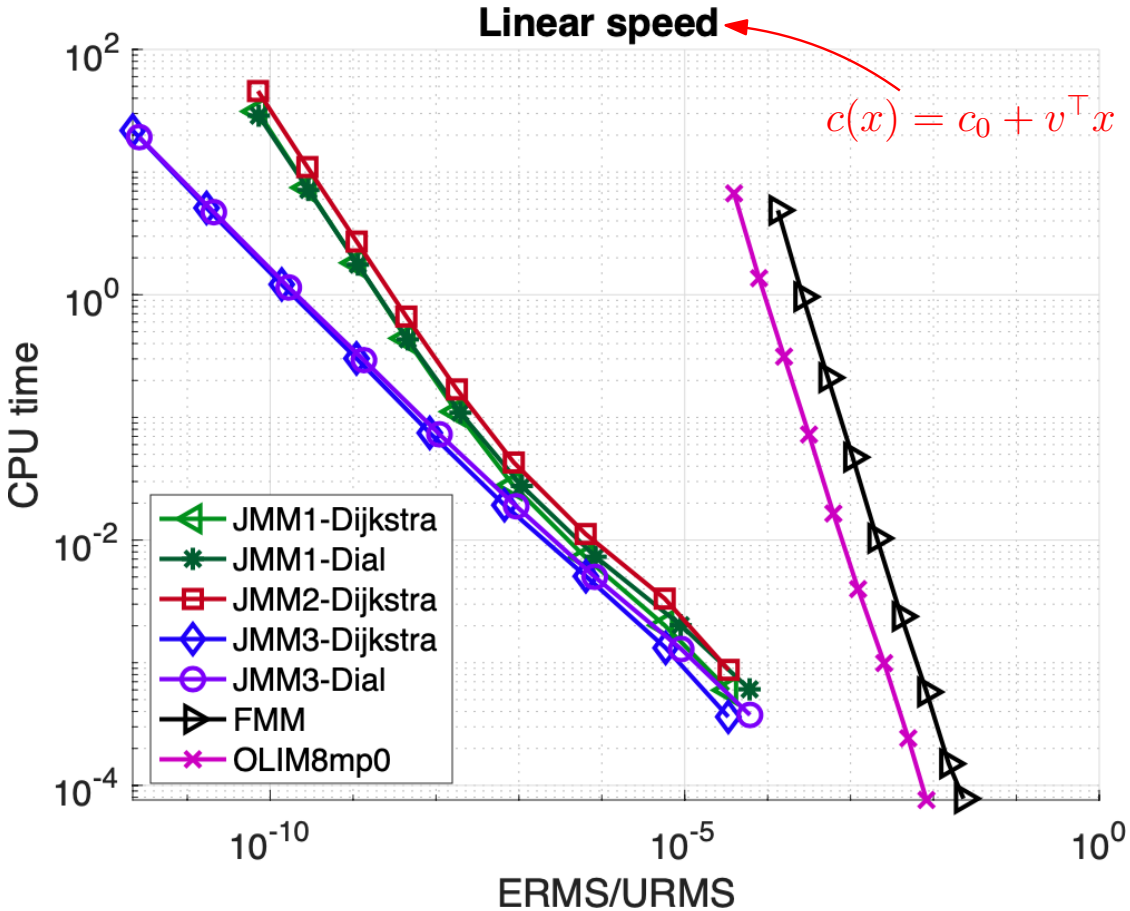

$$\begin{bmatrix} \dot{Q} \\ \dot{P} \end{bmatrix} = \begin{bmatrix} & c_0 I \\ \frac{-1}{c_0} \frac{\partial^2 c}{\partial q \partial q^\top}\Big|_{q=0} & \end{bmatrix} \begin{bmatrix} Q \\ P \end{bmatrix}$$

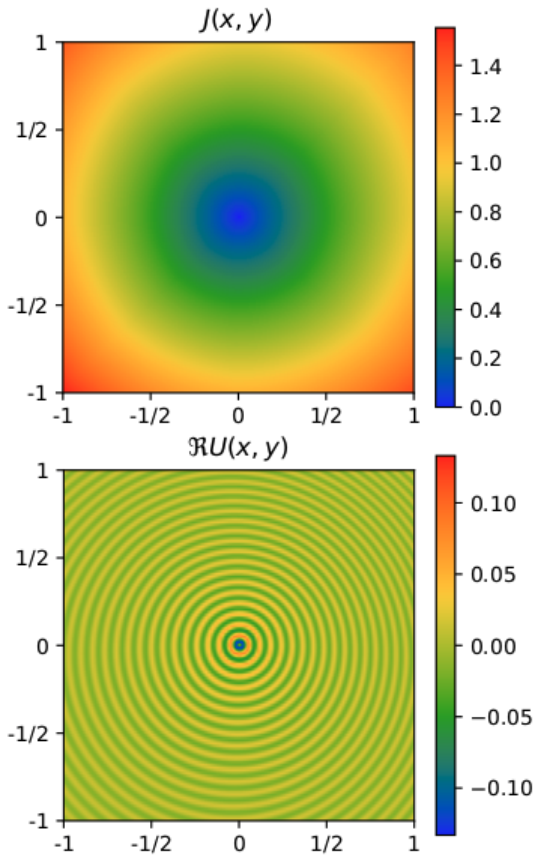
$$J = |\det Q|, \quad A(\phi_0(\sigma)) = \sqrt{\frac{c(\phi_0(\sigma))}{J(\phi_0(\sigma))}} A(\phi_0(0))$$




Potter and Cameron, 2021.

# 2D JMMs: numerical results

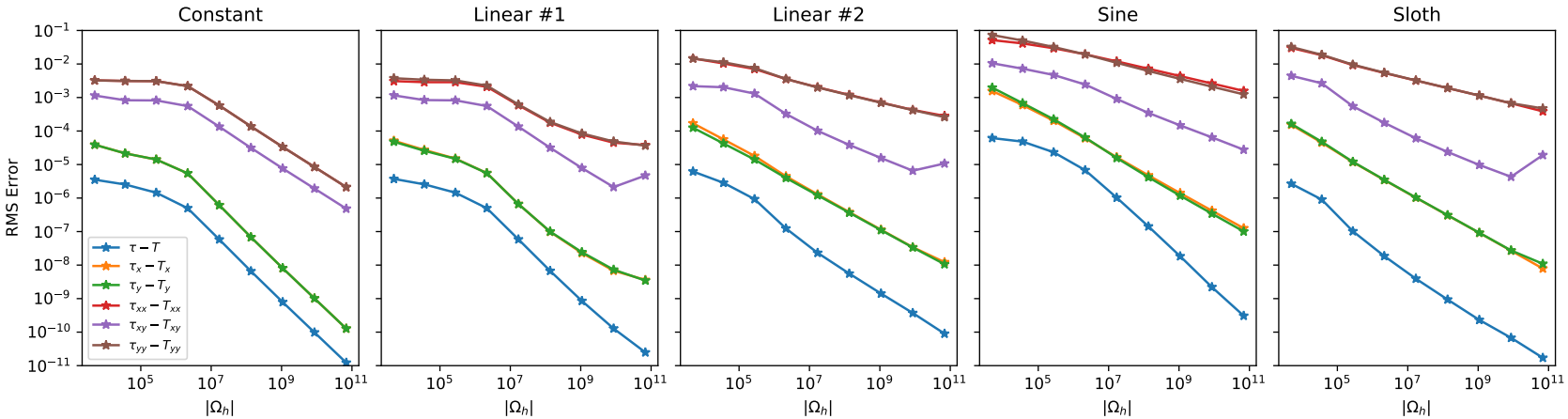



Potter and Cameron, 2021.

# 2D JMMs: numerical results

Results using cell marching:




Order of convergence ($O(h^p)$) for jets:

| | $\tau - T$ | $\tau_x - T_x$ | $\tau_y - T_y$ | $\tau_{xx} - T_{xx}$ | $\tau_{xy} - T_{xy}$ | $\tau_{yy} - T_{yy}$ |
|---|---|---|---|---|---|---|
| Constant | 3.09 | 3.11 | 3.11 | 2.01 | 2.05 | 2.01 |
| Linear #1 | 2.99 | 2.43 | 2.40 | 1.39 | 2.01 | 1.39 |
| Linear #2 | 2.10 | 1.76 | 1.72 | 0.77 | 1.25 | 0.77 |
| Sine | 2.91 | 1.80 | 1.89 | 0.73 | 1.31 | 0.80 |
| Sloth | 2.03 | 1.76 | 1.75 | 0.75 | 1.33 | 0.76 |

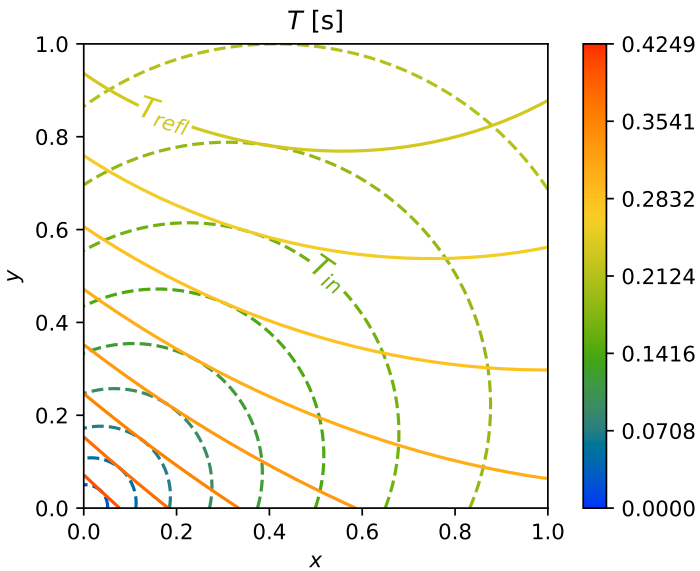
Potter and Cameron, 2021.

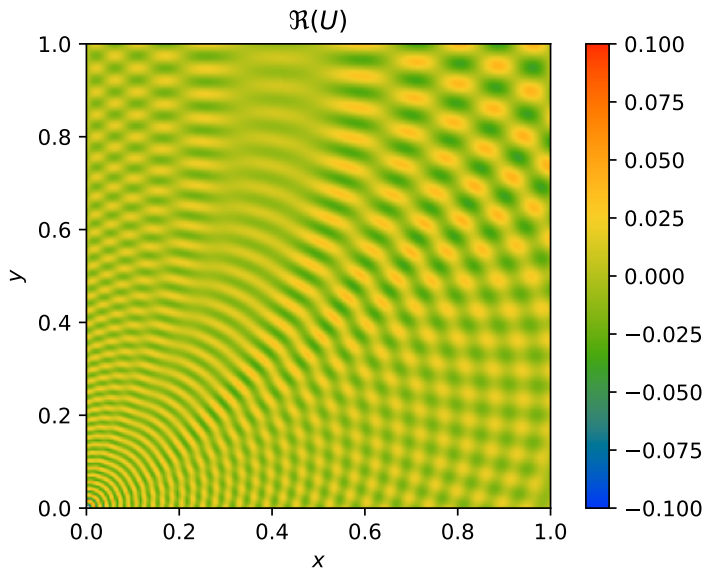
# 2D JMMs: reflections

Boundary conditions for reflected problem:

$$\tau_{\text{in}} = \tau_{\text{out}}, \quad J_{\text{in}} = J_{\text{out}} \quad \text{on wall}$$

and reflect $\nabla\tau_{\text{in}}$ over wall to get $\nabla\tau_{\text{out}}$:



# A JMM for an unstructured tetrahedron mesh

# 3D JMM: setup

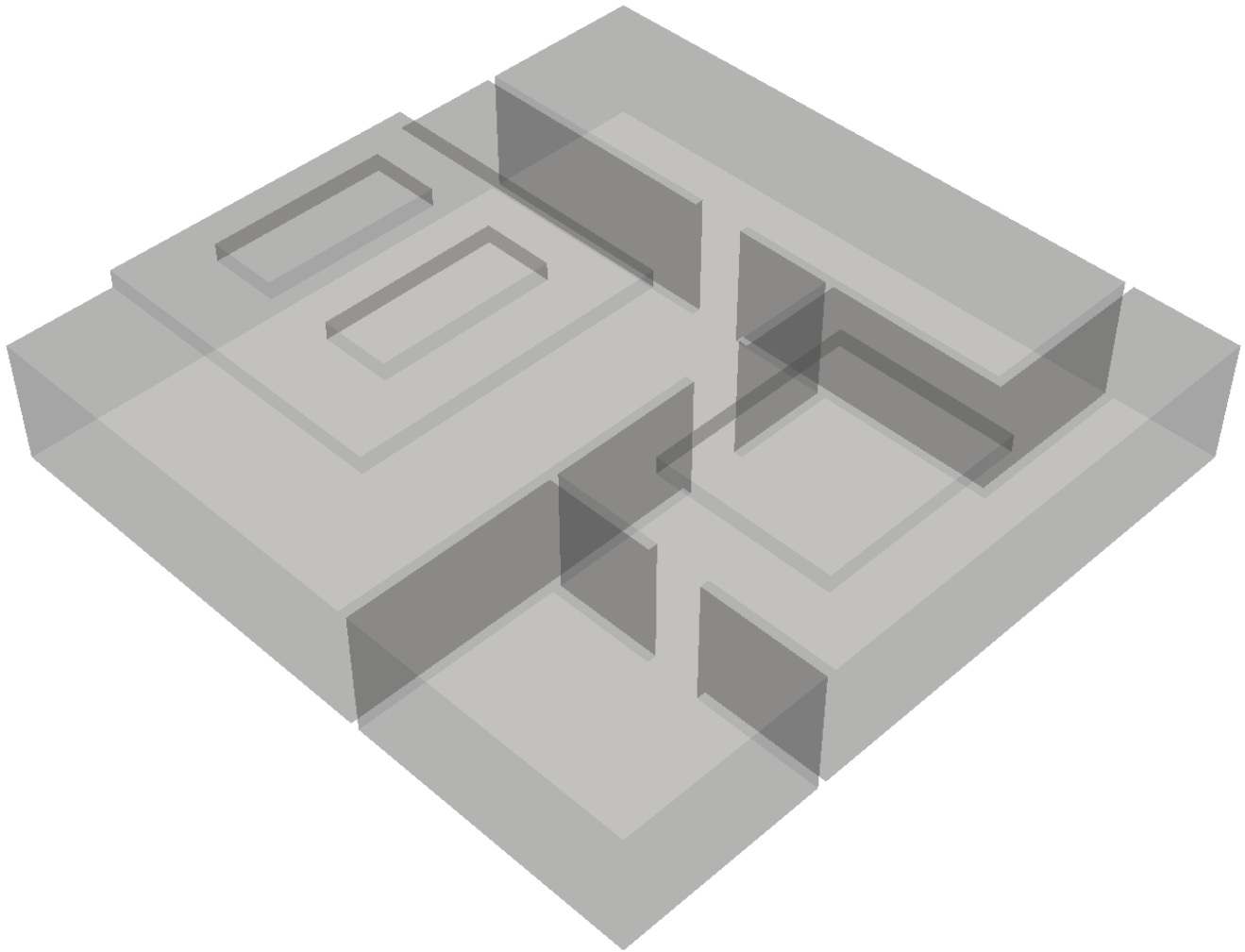
Develop a JMM to solve the eikonal equation on a tetrahedron mesh in 3D

**Goals**: $O(N \log N)$ solver with at least $O(h^2)$ accurate $T$ and $\nabla T$, so that we can compute $\alpha$ consistently

**Desiderata**: compact stencil, unstructured mesh, fast

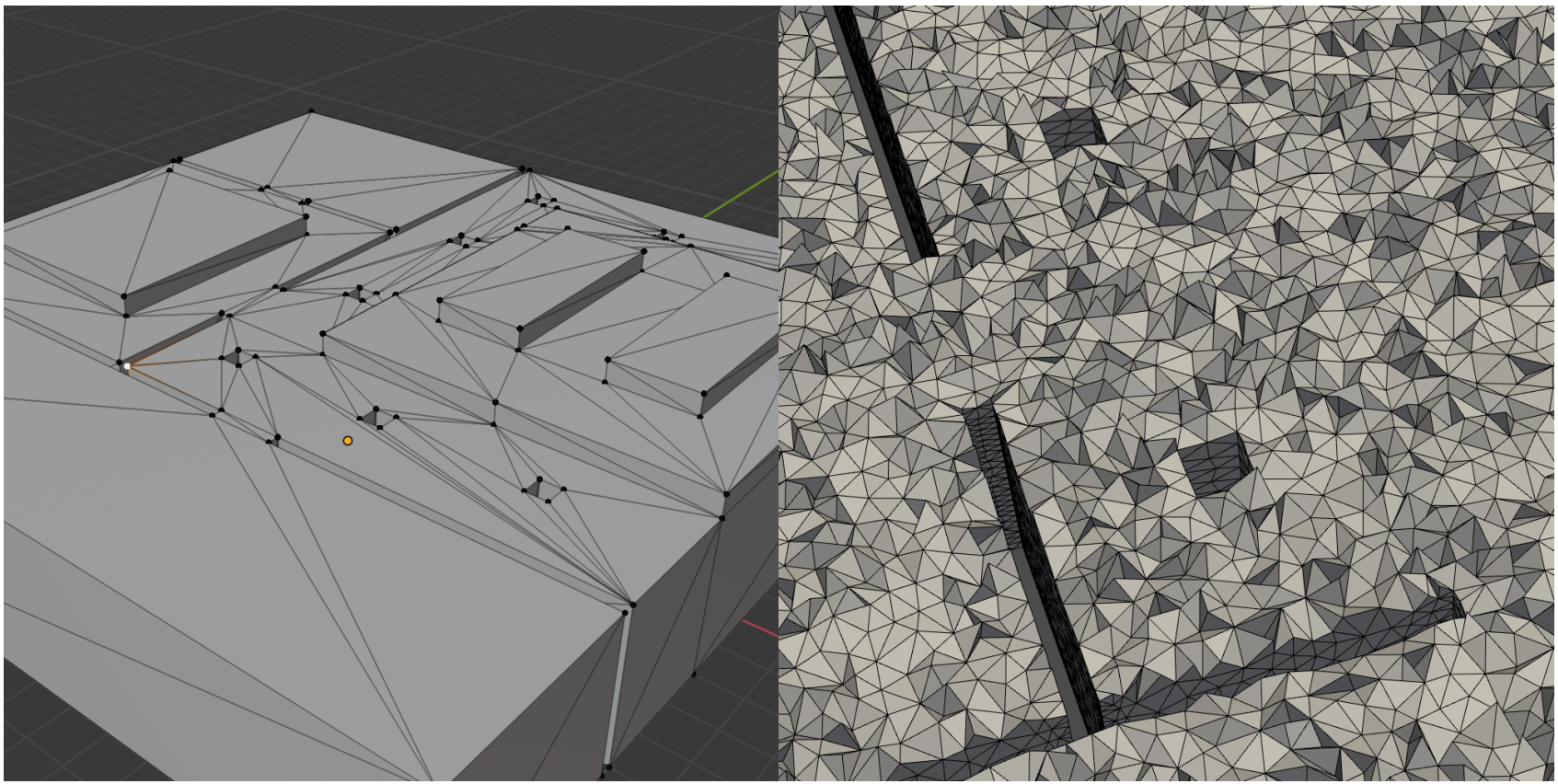
# 3D JMM: building test problem

Realistic test problem which includes some common architectural features, but not fully general (no small scale features, no curved surfaces, no floating obstacles, ...):



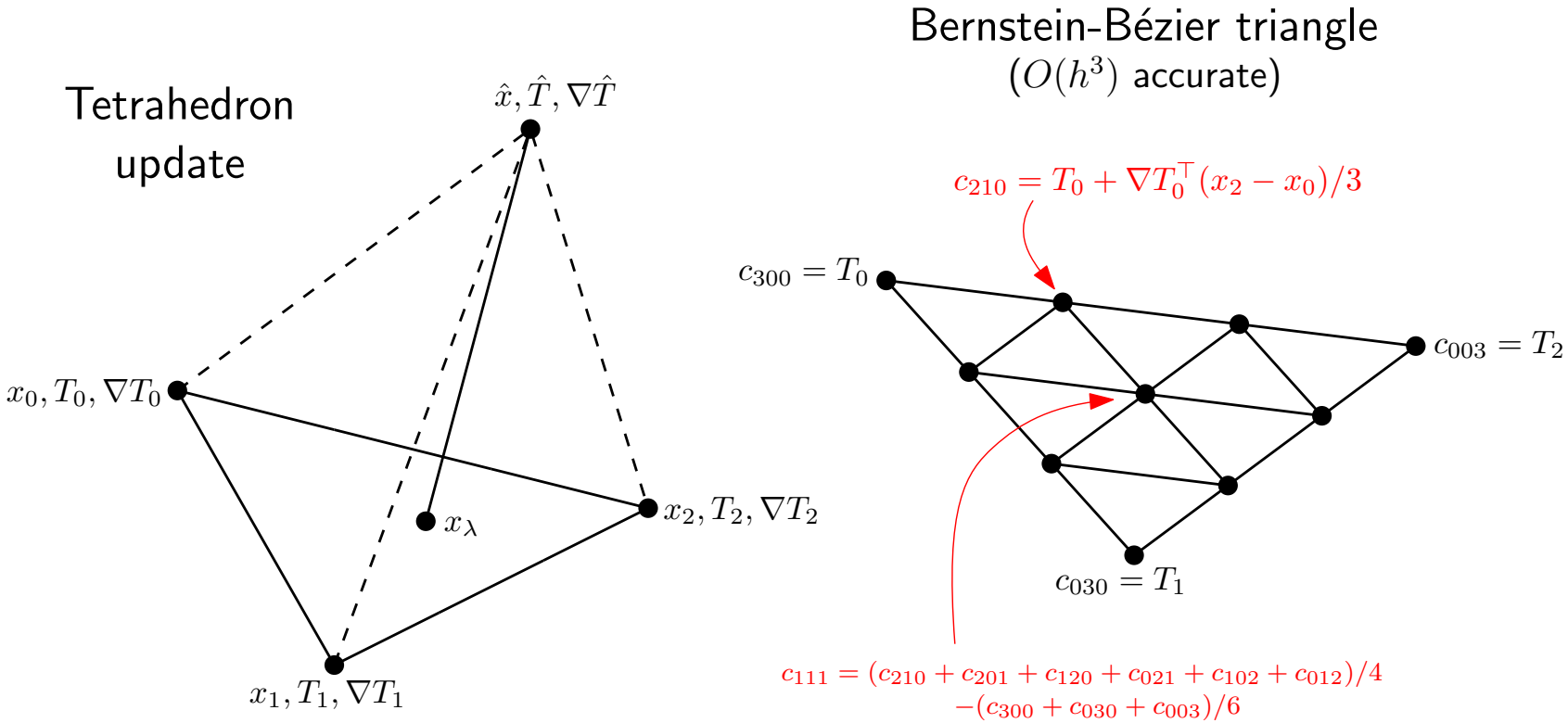
# 3D JMM: tetrahedron meshing

Use TetGen to mesh piecewise linear complex:



# 3D JMM: tetrahedron updates for $c = 1$

Workhorse for this solver is a cubically accurate tetrahedron update. Since $c \equiv 1$, rays are straight lines:

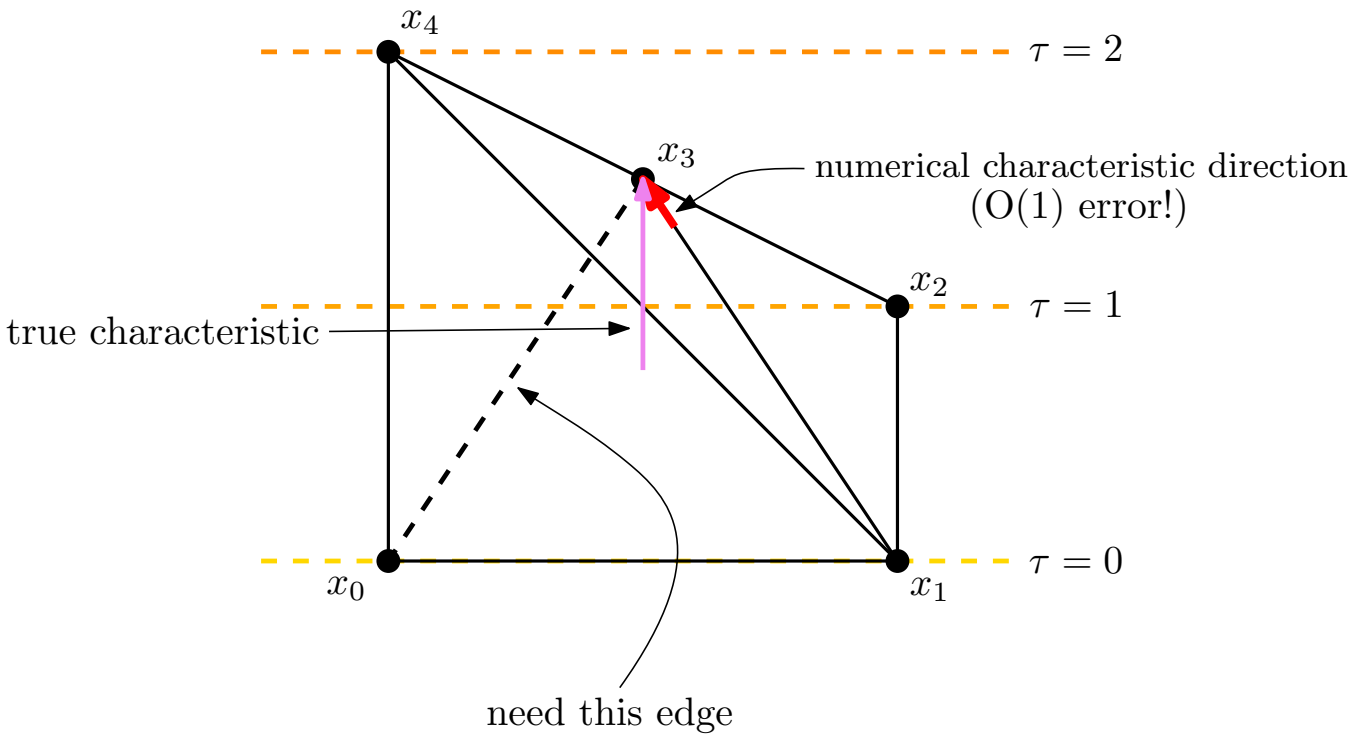



Minimize the cost function:

$$T(\hat{\boldsymbol{x}}) = \min_{\boldsymbol{\lambda}} \left\{ T(\boldsymbol{x}_{\boldsymbol{\lambda}}) + \|\hat{\boldsymbol{x}} + \boldsymbol{x}_{\boldsymbol{\lambda}}\| \right\}$$

# 3D JMM: causality

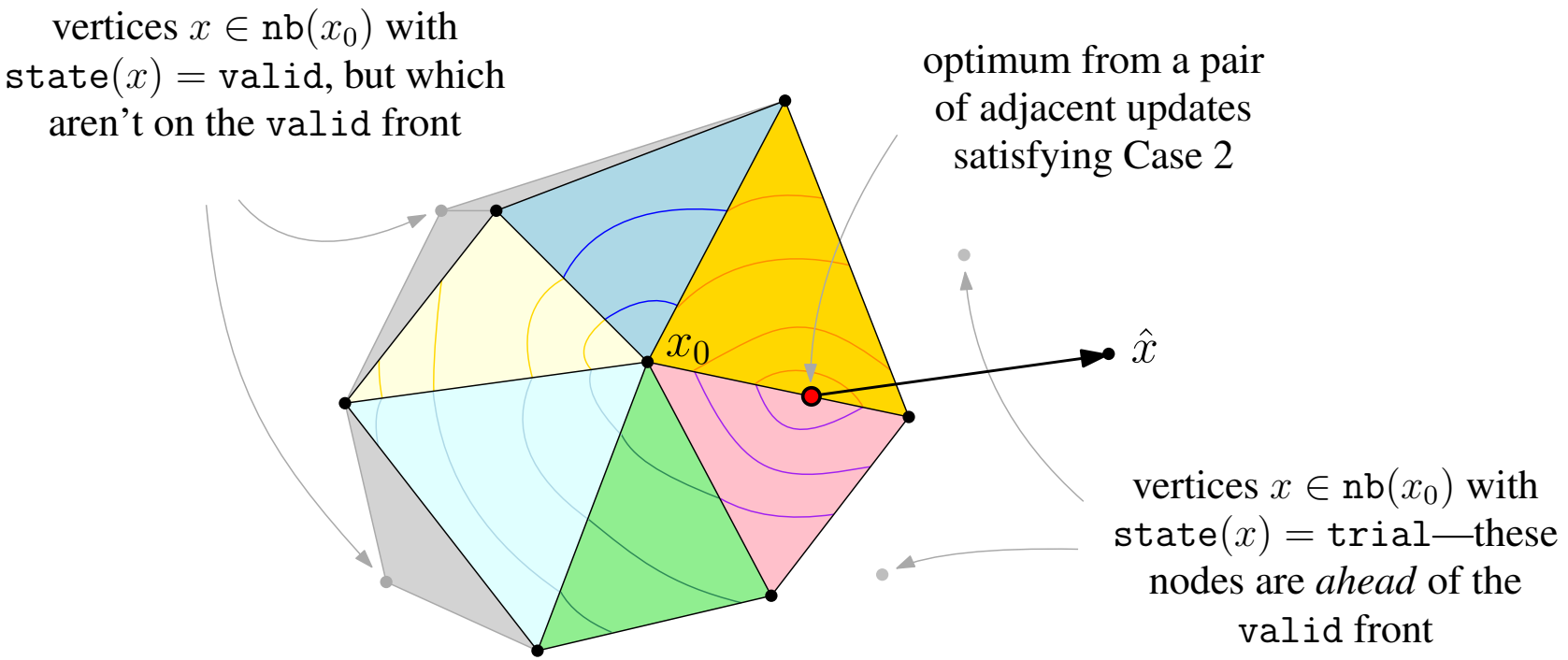
We can't use the topology of the tetrahedron mesh to define update neighborhoods:




Prior works on causal marchers on triangle meshes by Kimmel, Sethian, and Vladimirsky.

# 3D JMM: enforcing causality

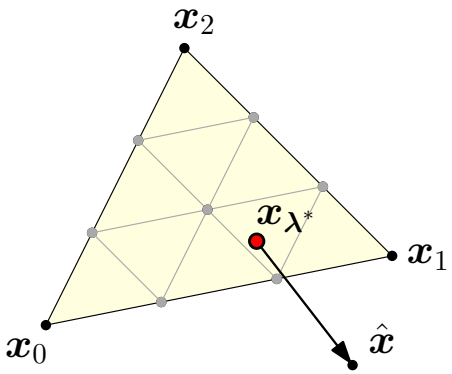
We want to find the point $\boldsymbol{x}_{\boldsymbol{\lambda}^*}$ which minimizes the cost function over each `valid` triangle on the `valid` front



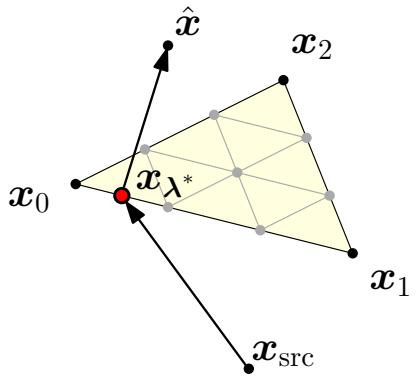
# 3D JMM: enforcing causality

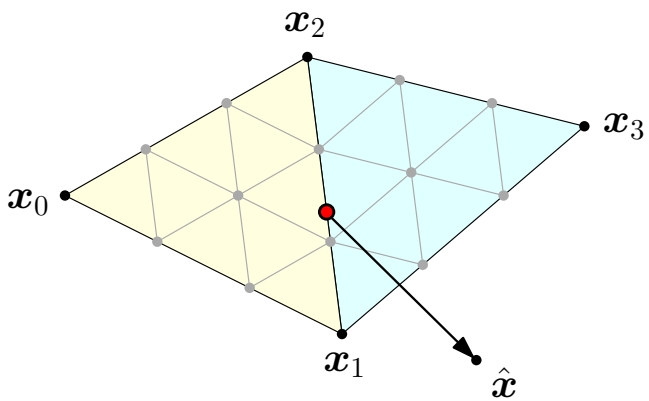
Check the Lagrange multipliers of a minimizer to see if the update is causal:

Interior point minimizer

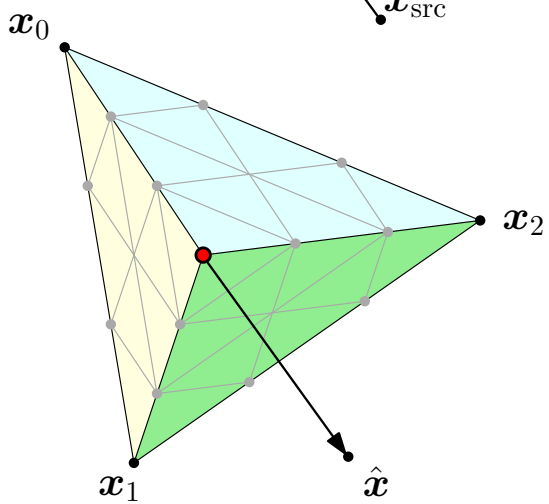



Unphysical ray



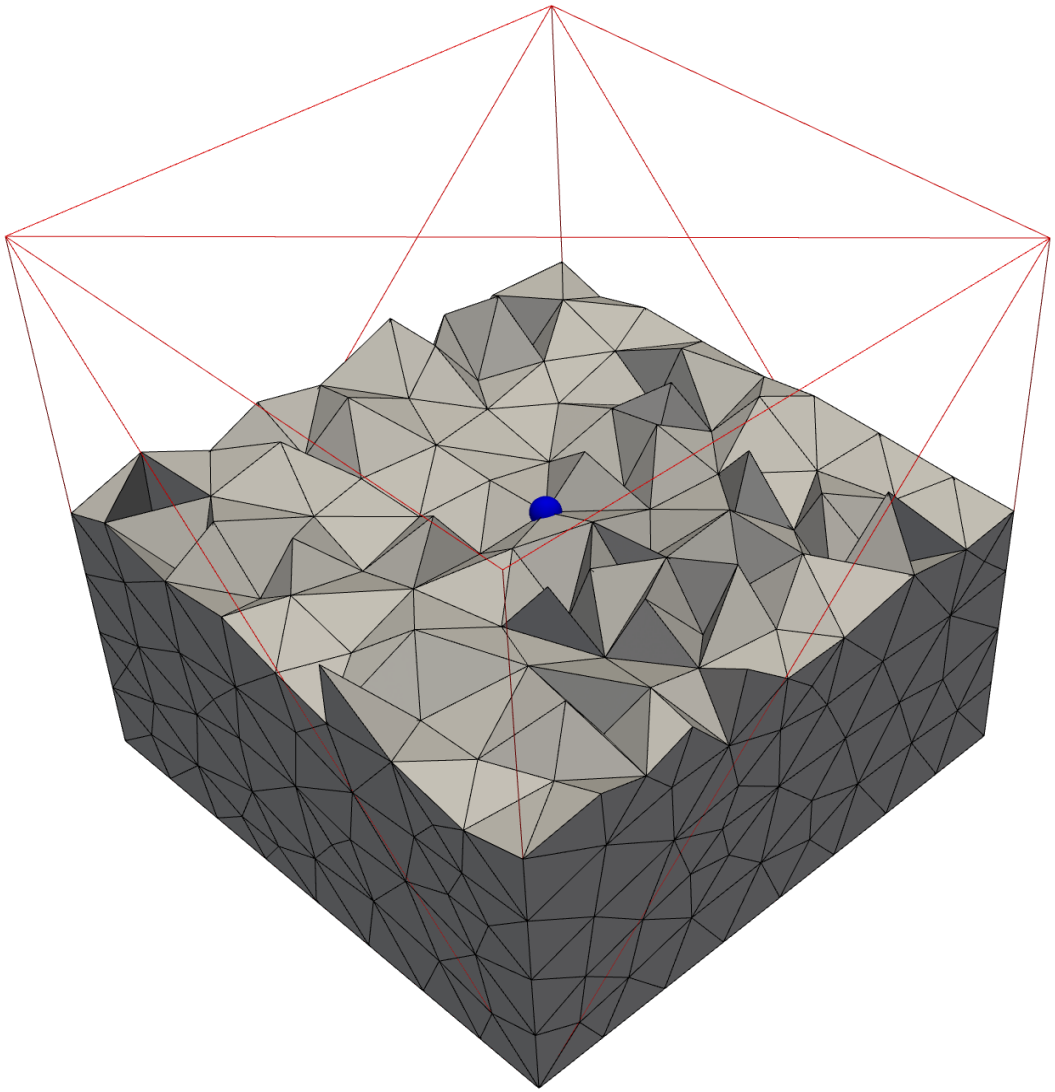




Boundary minimizer from two updates




. . . and from three updates

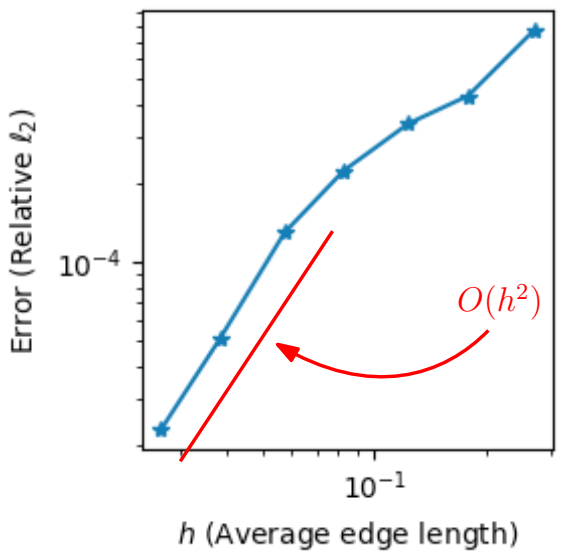
# 3D JMM: testing on a box domain



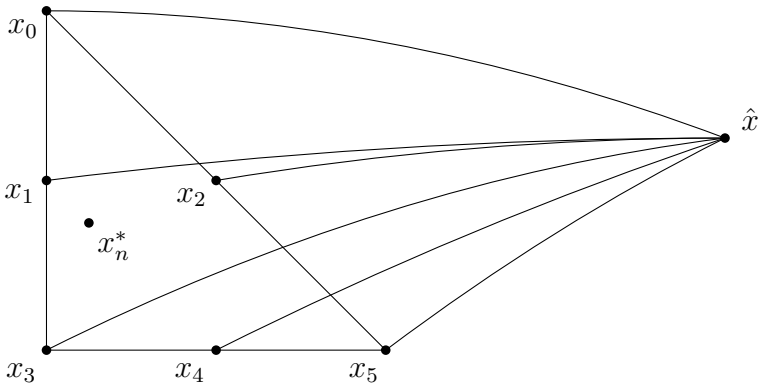
▲Input mesh

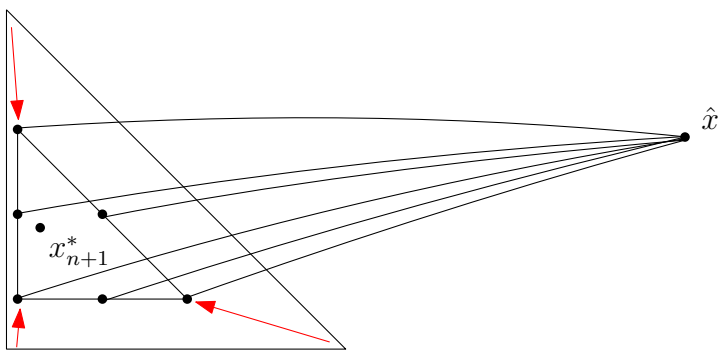
▲Tetrahedron mesh

▲Point source



# 3D JMM: $c \neq 1$




For $c \neq 1$, cost function is more complex, SQP is annoying because gradient and Hessian are complicated

Instead, use "line update" (two-point BVP for raytracing) as building block

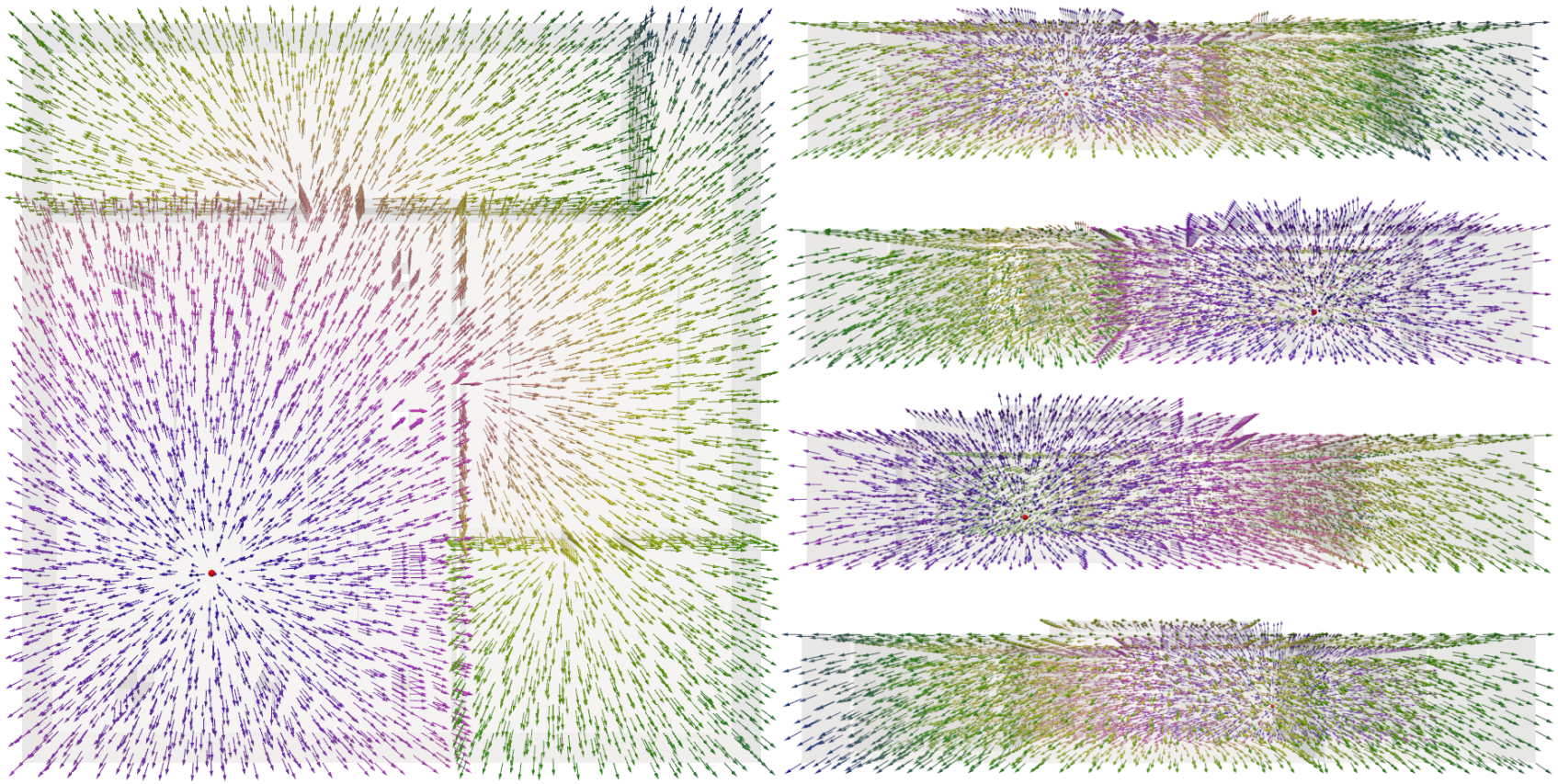
Solve 6 line updates at nodes of quadratic Lagrange triangle

Solve inequality constrainted QP

Shrink triangle towards optimum

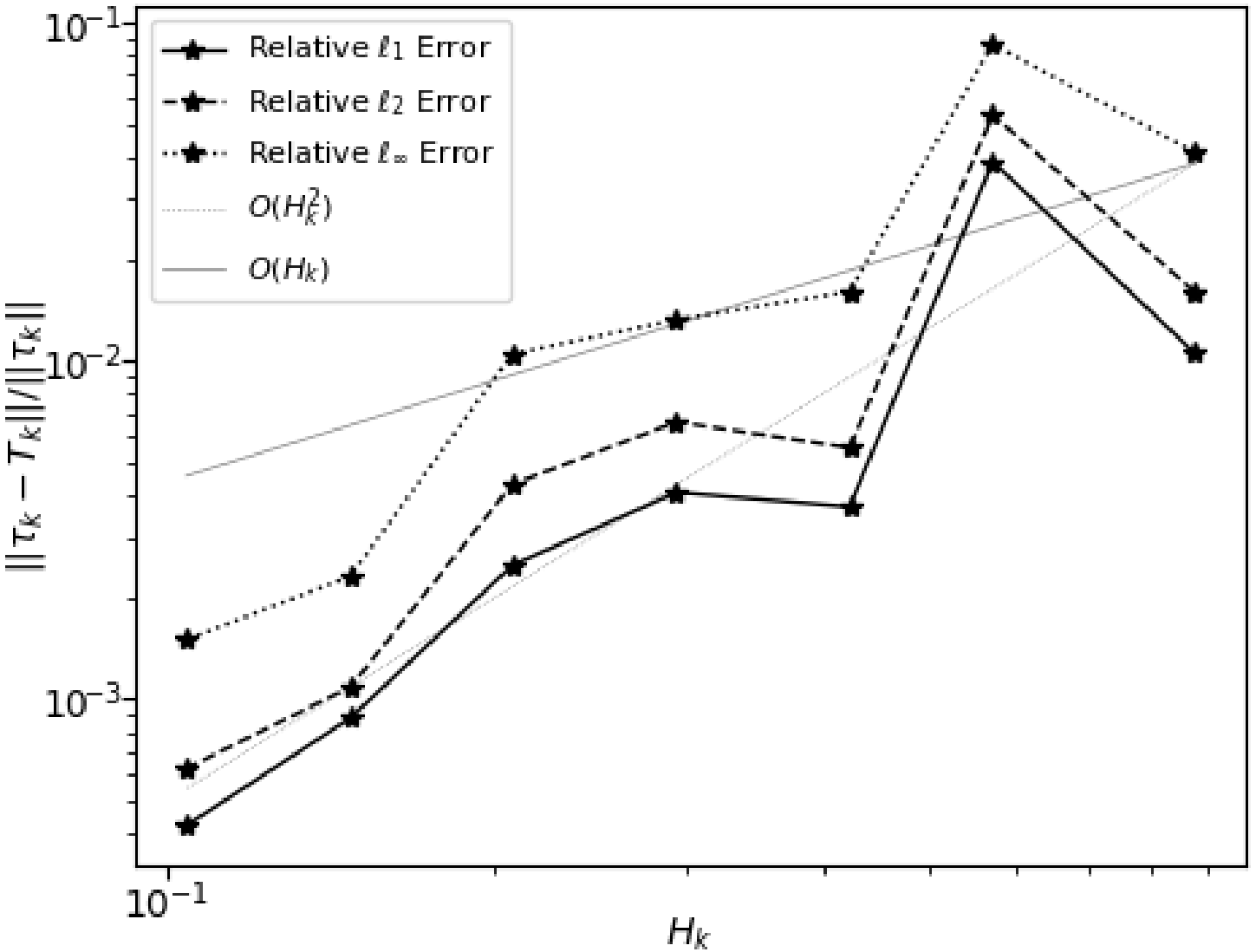
# 3D JMM: $c \neq 1$

Solution in test building geometry (arrows: $\nabla T$, colors: $T$):



# 3D JMM: $c \neq 1$

Results for study on unit box with point source at origin:



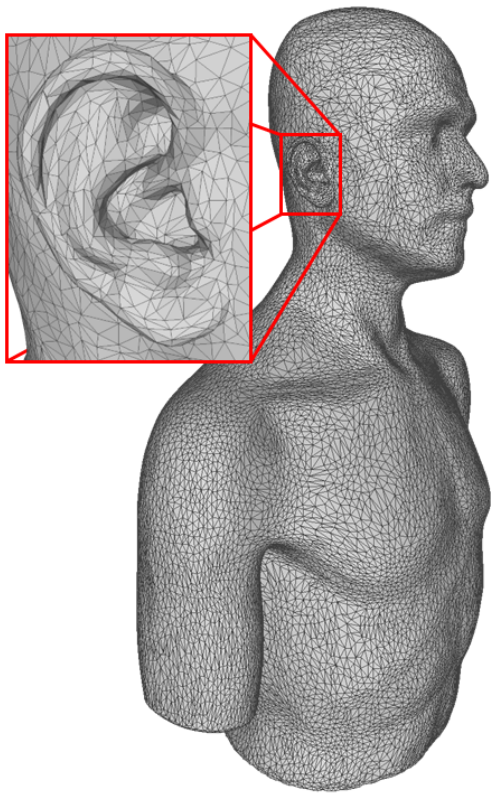
# 3D JMM: computing ITDs

Interaural time differences (ITDs) are a basic psychoacoustic quantity

Useful to simulate accurate in spatial audio (HRIR filter design)

Triangle mesh conforming to head + torso

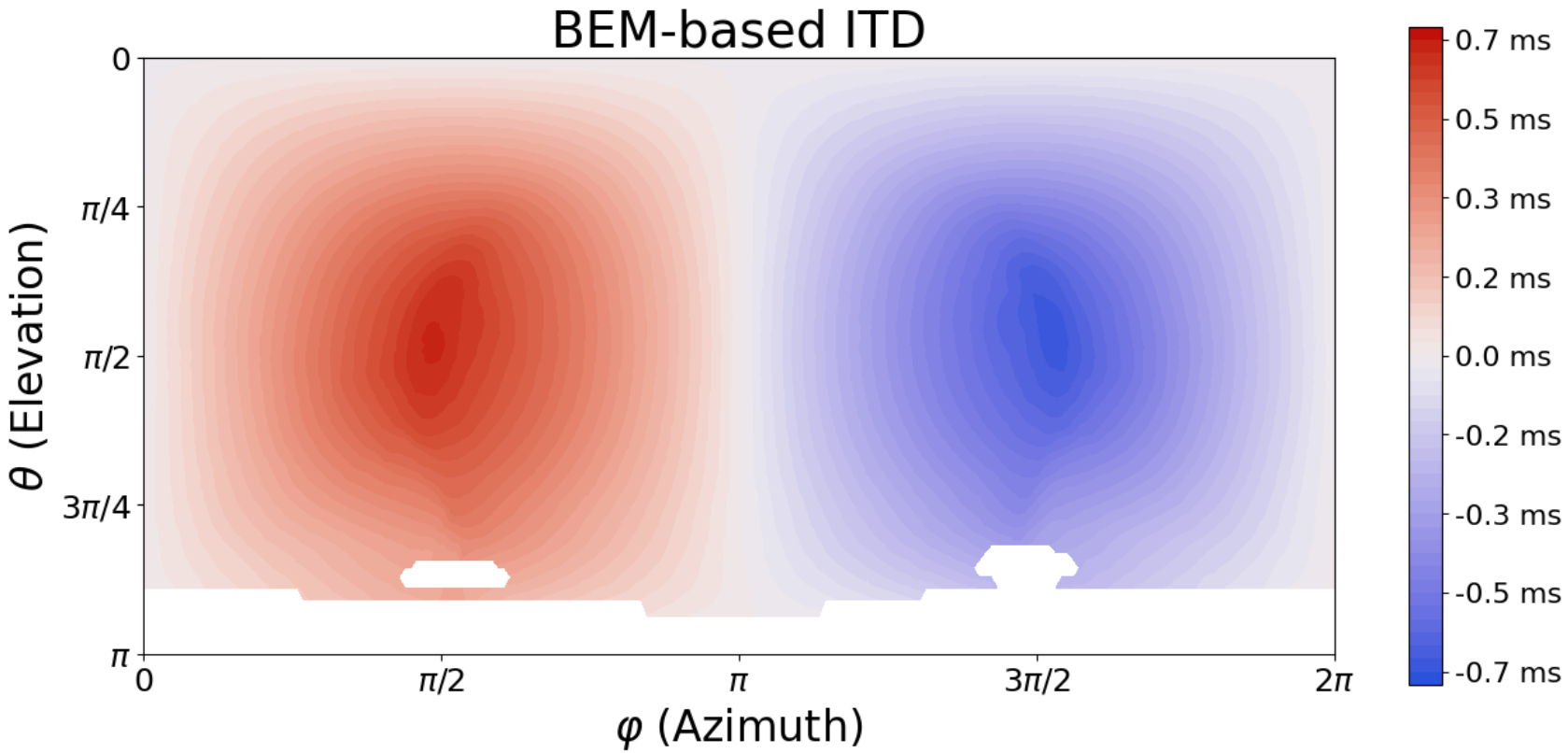
Place point sources in each ear canal, simulate response and compute differences



P, Hoover, Zotkin, Duraiswami, 2023.

# 3D JMM: computing ITDs

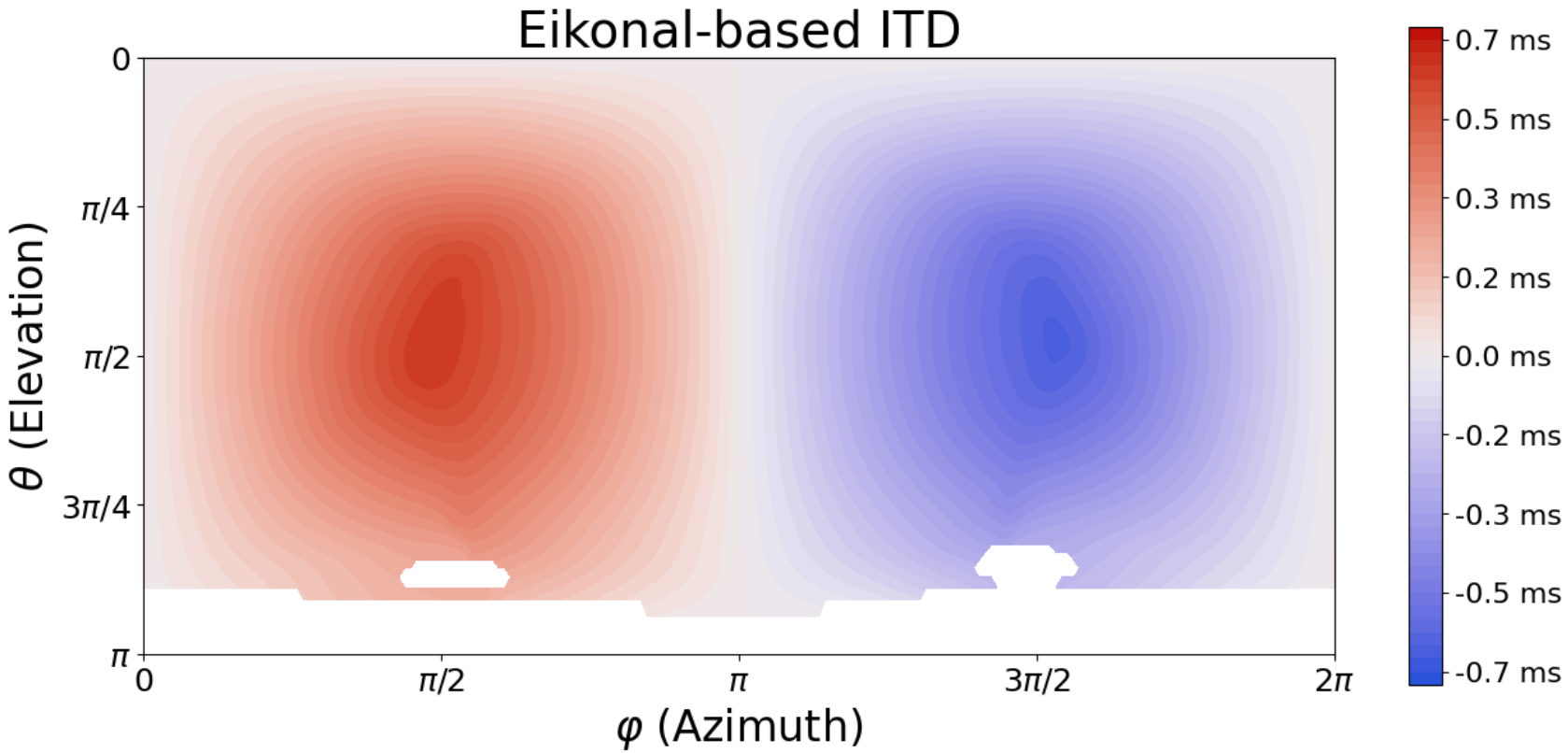
Solve Helmholtz equation on uniform frequency grid using FMM-accelerated boundary element method. IFFT result and use filter to estimate onset.




P, Hoover, Zotkin, Duraiswami, 2023.

# 3D JMM: computing ITDs

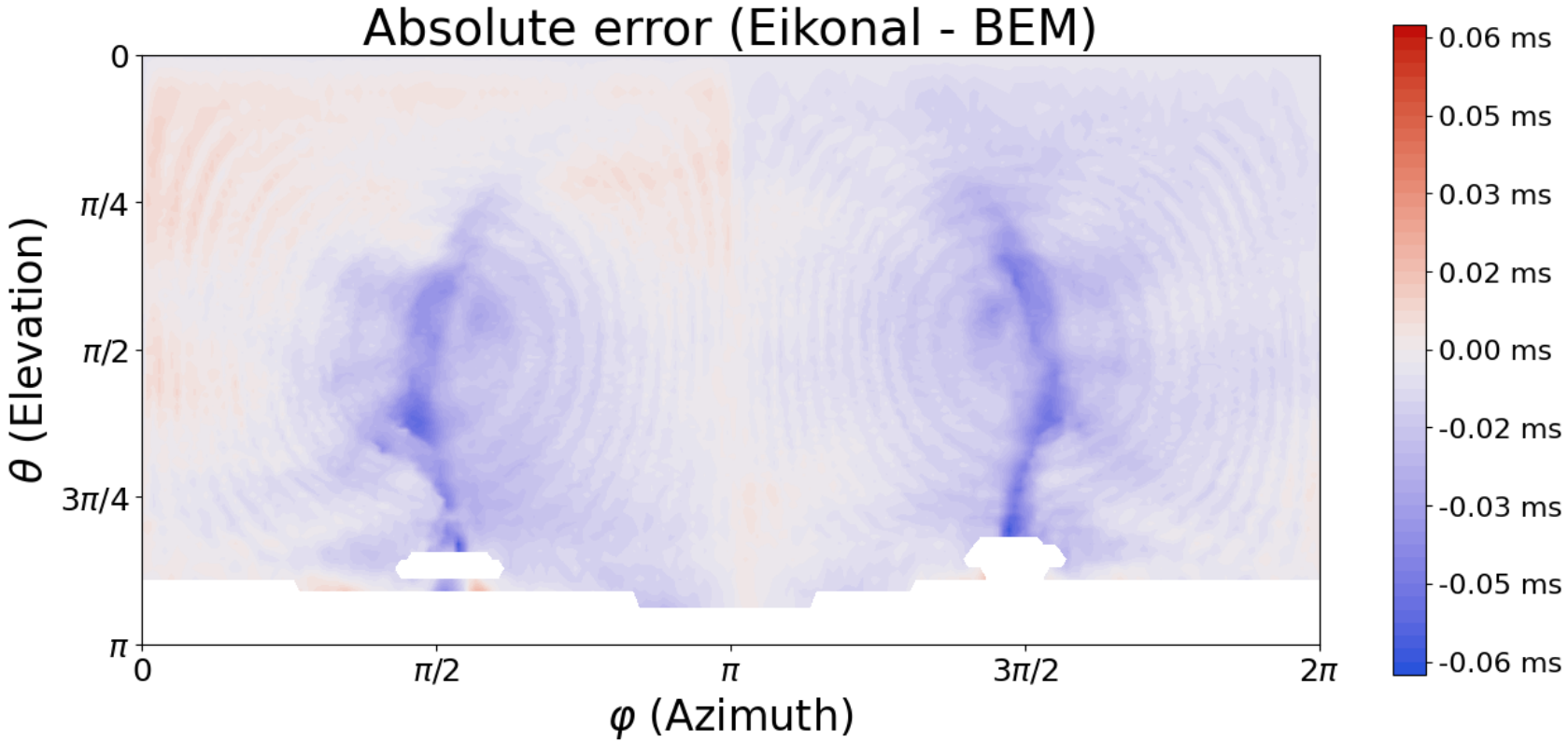
Compute ITD using Helmholtz reciprocity: difference of arrival times on sphere of some radius surrounding head




P, Hoover, Zotkin, Duraiswami, 2023.

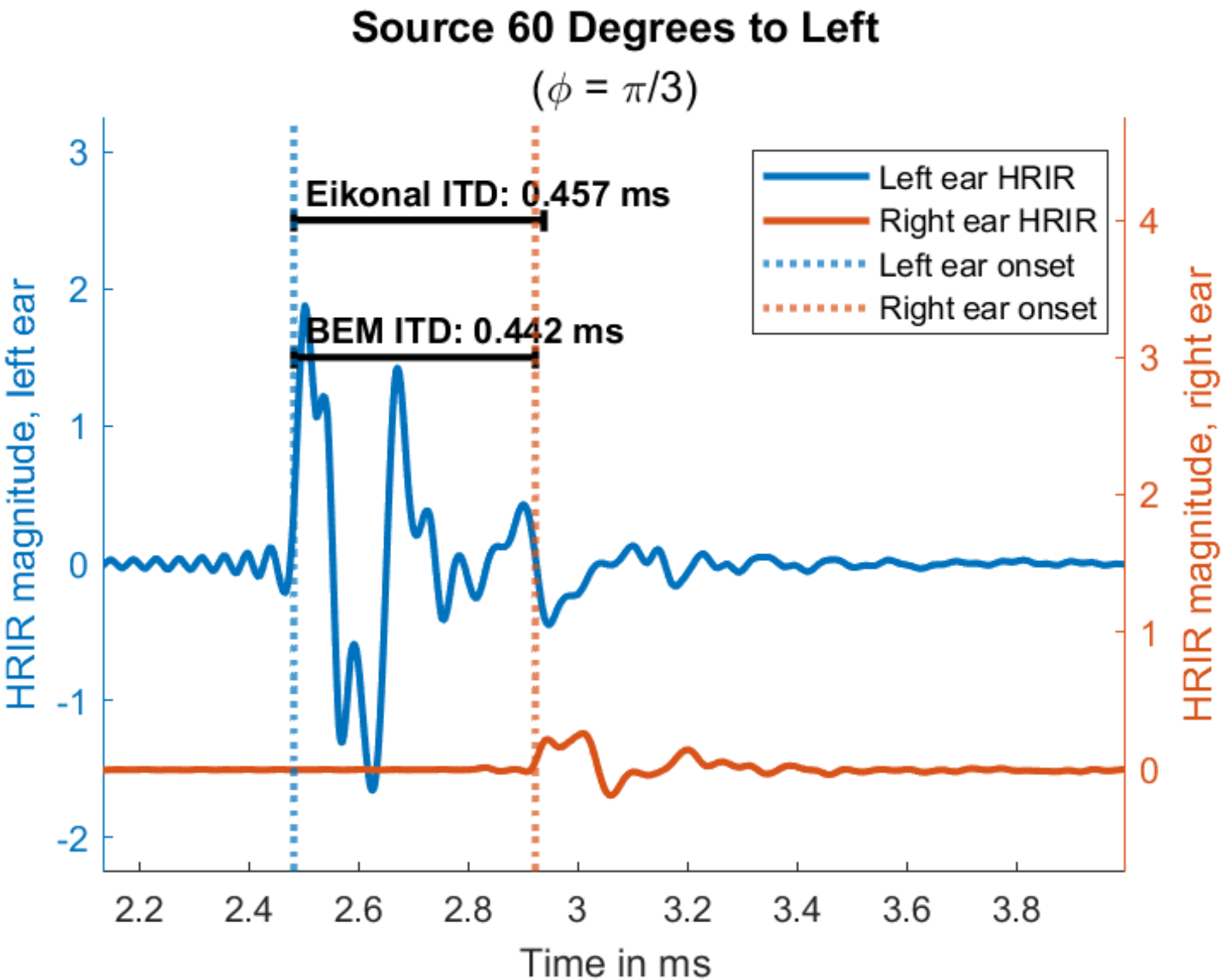
# 3D JMM: computing ITDs

Error is on the order of 1 JND ("just noticeable difference"):




P, Hoover, Zotkin, Duraiswami, 2023.

# 3D JMM: computing ITDs

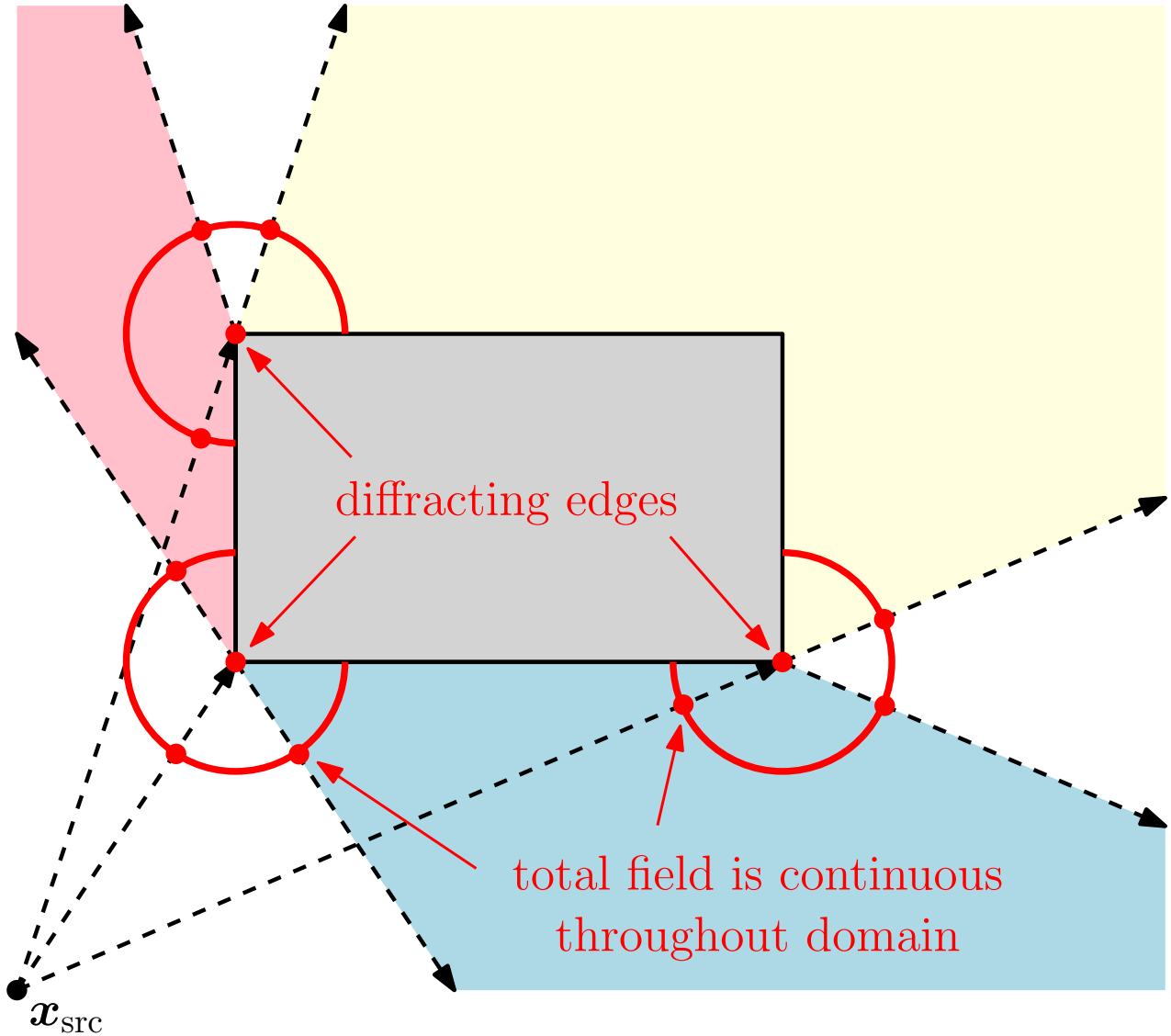



P, Hoover, Zotkin, Duraiswami, 2023.

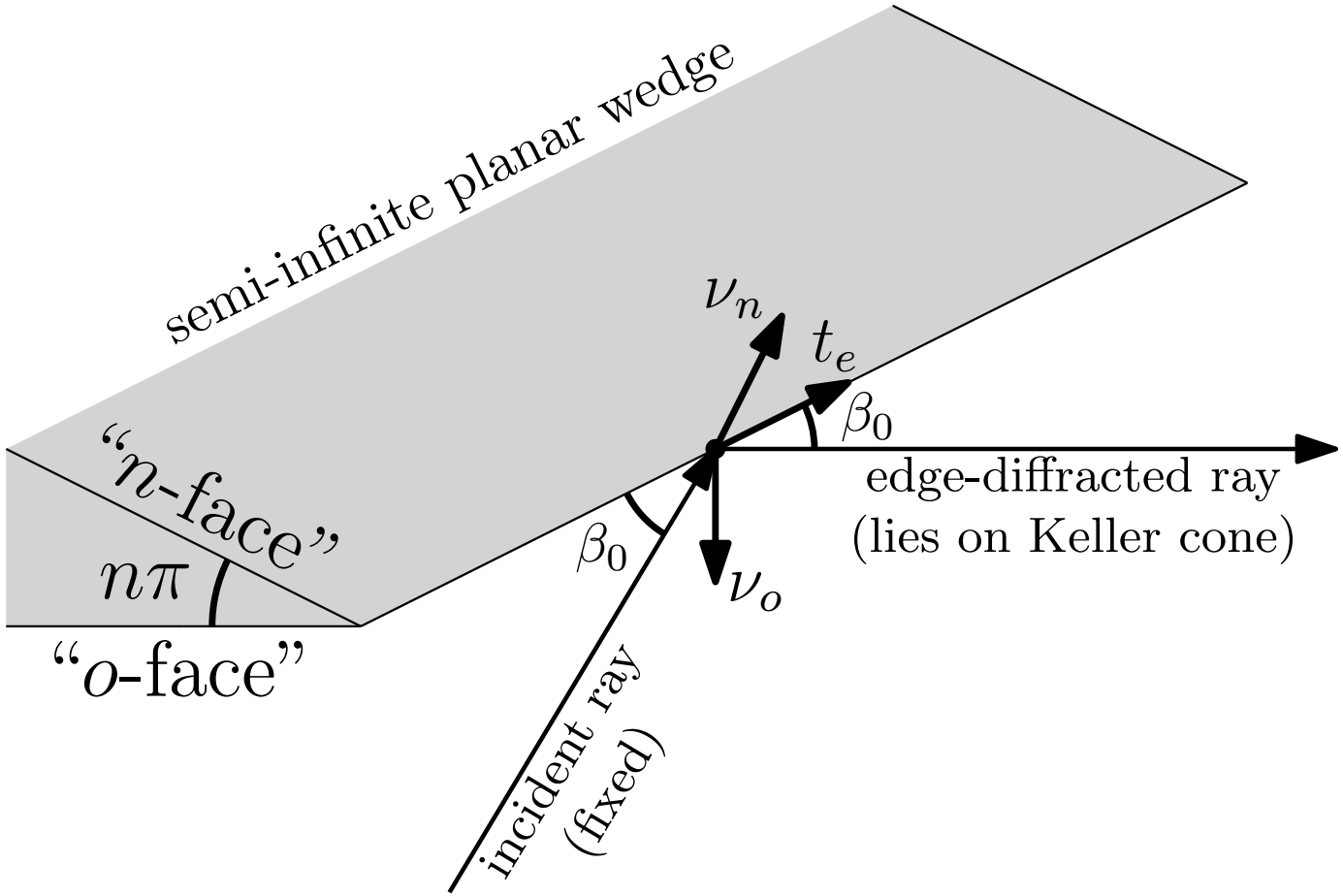
# Multiple arrivals in 3D

# MA in 3D: multiple arrivals

Multiple arrivals: $u_\omega(\boldsymbol{x}) \sim \sum_{\substack{k\text{th ray} \\ \text{through } \boldsymbol{x}}} A_\omega^{(k)}(\boldsymbol{x}) e^{i\omega\tau^{(k)}(\boldsymbol{x})}$



# MA in 3D: edge diffraction




Generalized Fermat's principle with boundary constraints:

$$\min_{\boldsymbol{\psi}} \int_{\boldsymbol{\psi}} s(\boldsymbol{\psi}) d\ell \quad \text{s.t.} \quad \boldsymbol{\psi}_i \in \partial\Omega_i$$

# MA in 3D: BCs for reflected and diffracted eikonal

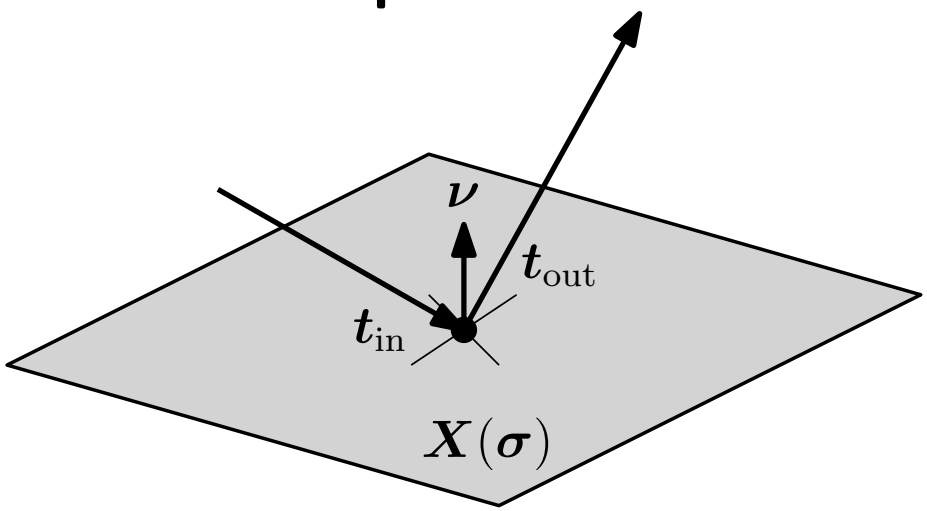
Boundary conditions for the reflected and diffracted eikonal:

$$\tau_{\text{in}}(\boldsymbol{X}(\boldsymbol{\sigma})) = \tau_{\text{out}}(\boldsymbol{X}(\boldsymbol{\sigma}))$$

Taking the gradient gives:

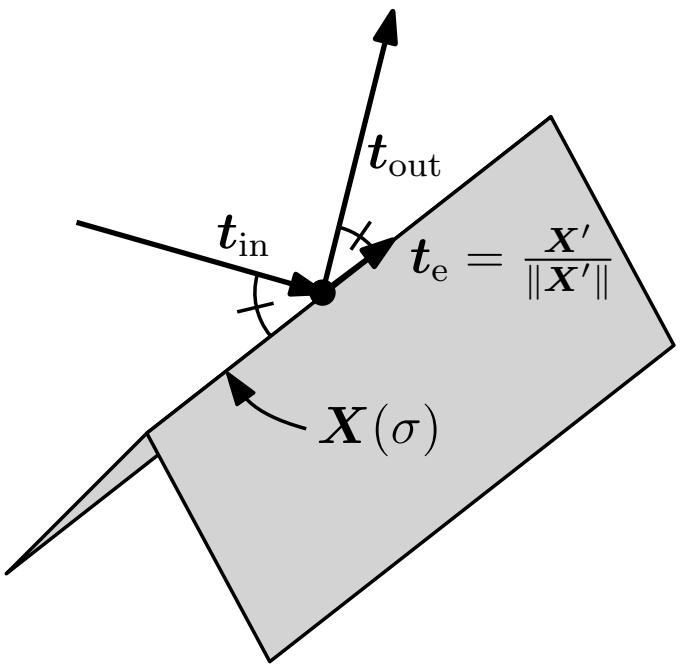
$$\nabla \boldsymbol{X}(\boldsymbol{\sigma})^\top \nabla \tau_{\text{in}}(\boldsymbol{X}(\boldsymbol{\sigma})) = \nabla \boldsymbol{X}(\boldsymbol{\sigma})^\top \nabla \tau_{\text{out}}(\boldsymbol{X}(\boldsymbol{\sigma}))$$

This leads to **specular reflection** and the **Keller cone**:



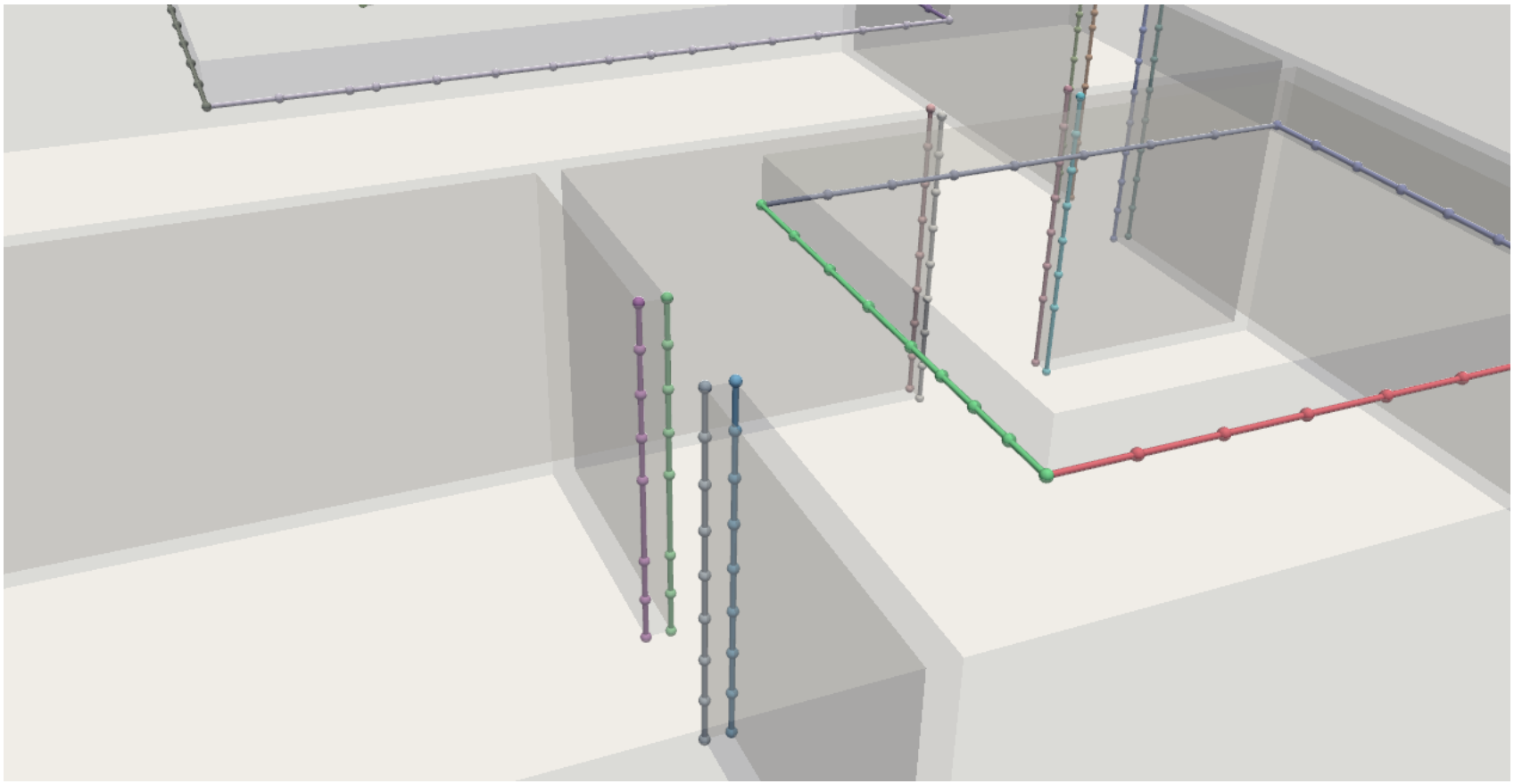

$$\boldsymbol{\nu} \in \text{null}(\boldsymbol{X})$$

$$\boldsymbol{t}_{\text{out}} = \left(\boldsymbol{I} - 2\boldsymbol{\nu}\boldsymbol{\nu}^\top\right)\boldsymbol{t}_{\text{in}}$$



# MA in 3D: reflectors and diffractors

Restrict $\tau_{\text{in}}$ to reflecting and diffracting subsets of $\partial\Omega$ and compute scattered fields:



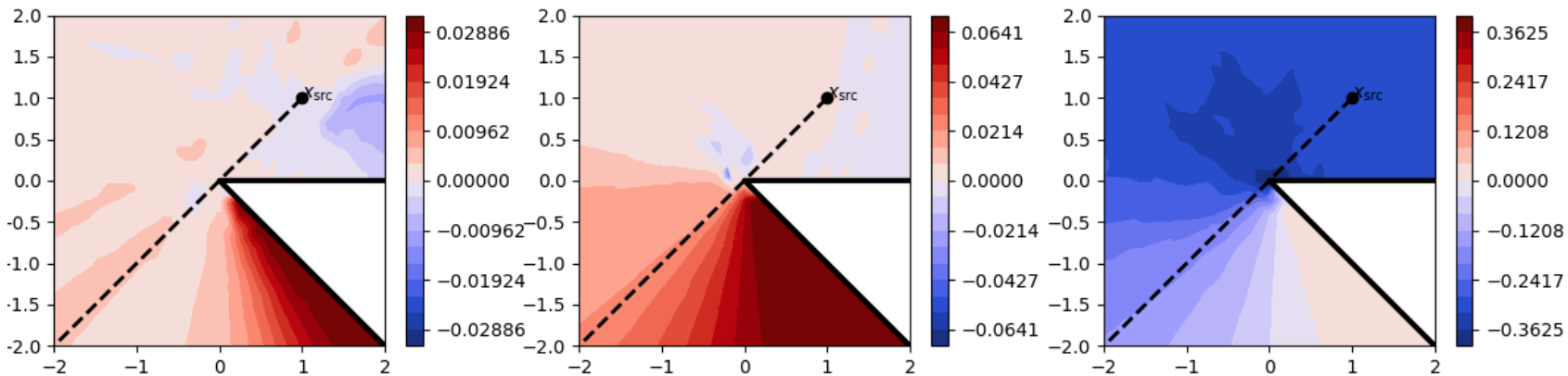
# MA in 3D: wedge test problem

Reflected and diffracted fields are hard to compute accurately

Need a careful reinitialization scheme (cf. locally factored eikonal equation)—ask me offline or read our paper if you want details

We achieve desired $O(h^2)$ accuracy on wedge problem

Cross-section of 3D error field (interpolate from tetrahedral Beziér spline to grid $\rightarrow$ imshow):



# MA in 3D: numerical shadows using `origin`

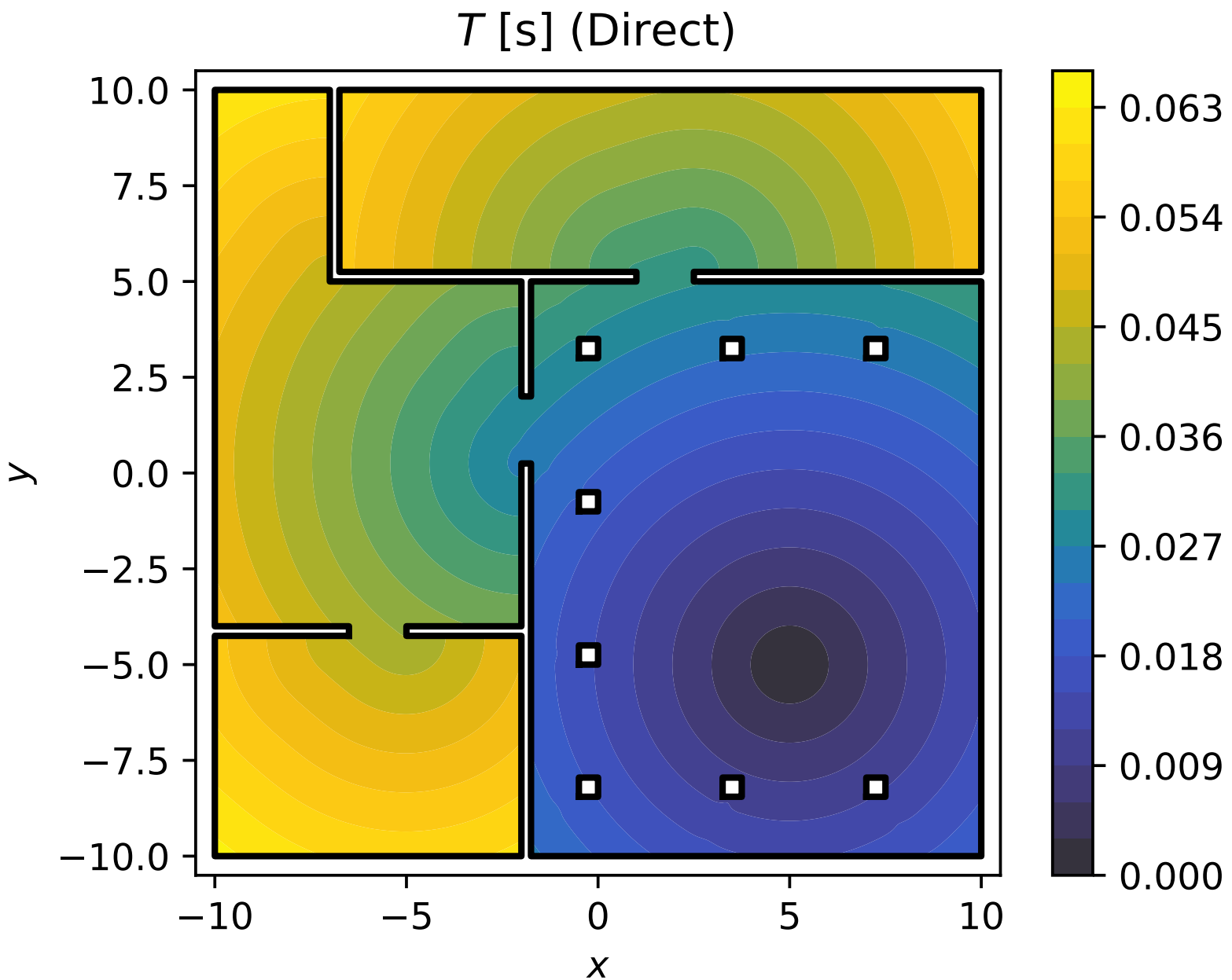
As we march, we record the "`parent`" of each node (nodes $x_0, x_1, x_2$ in final update & convex coefficients giving $x_{\lambda^*}$)

Afterwards, we can replay the dynamic programming plan to approximately propagate quantities along characteristics from boundary points by taking linear combinations

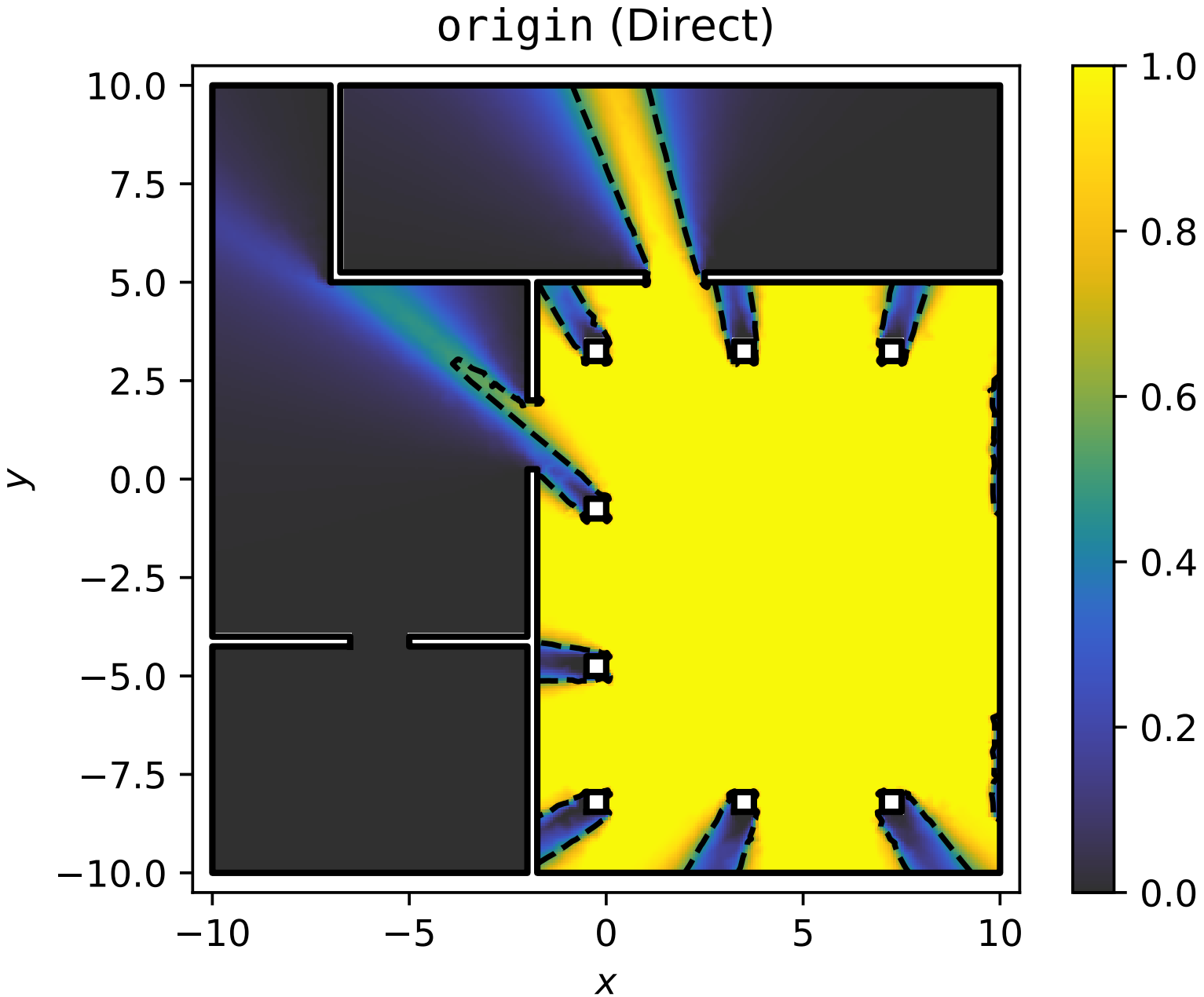
**Rough idea:** after solving, set $\texttt{origin}(x) = 0, \frac{1}{2}, 1$ if $x$ is shadowed, on the shadow boundary, or illuminated, respectively—also need to set `origin` for nodes in diffracting edges. Then, propagate `origin` field by replaying DP plan

**Goal:** Shadow boundary corresponds to $\frac{1}{2}$-level set of `origin`

# MA in 3D: numerical shadows using `origin`

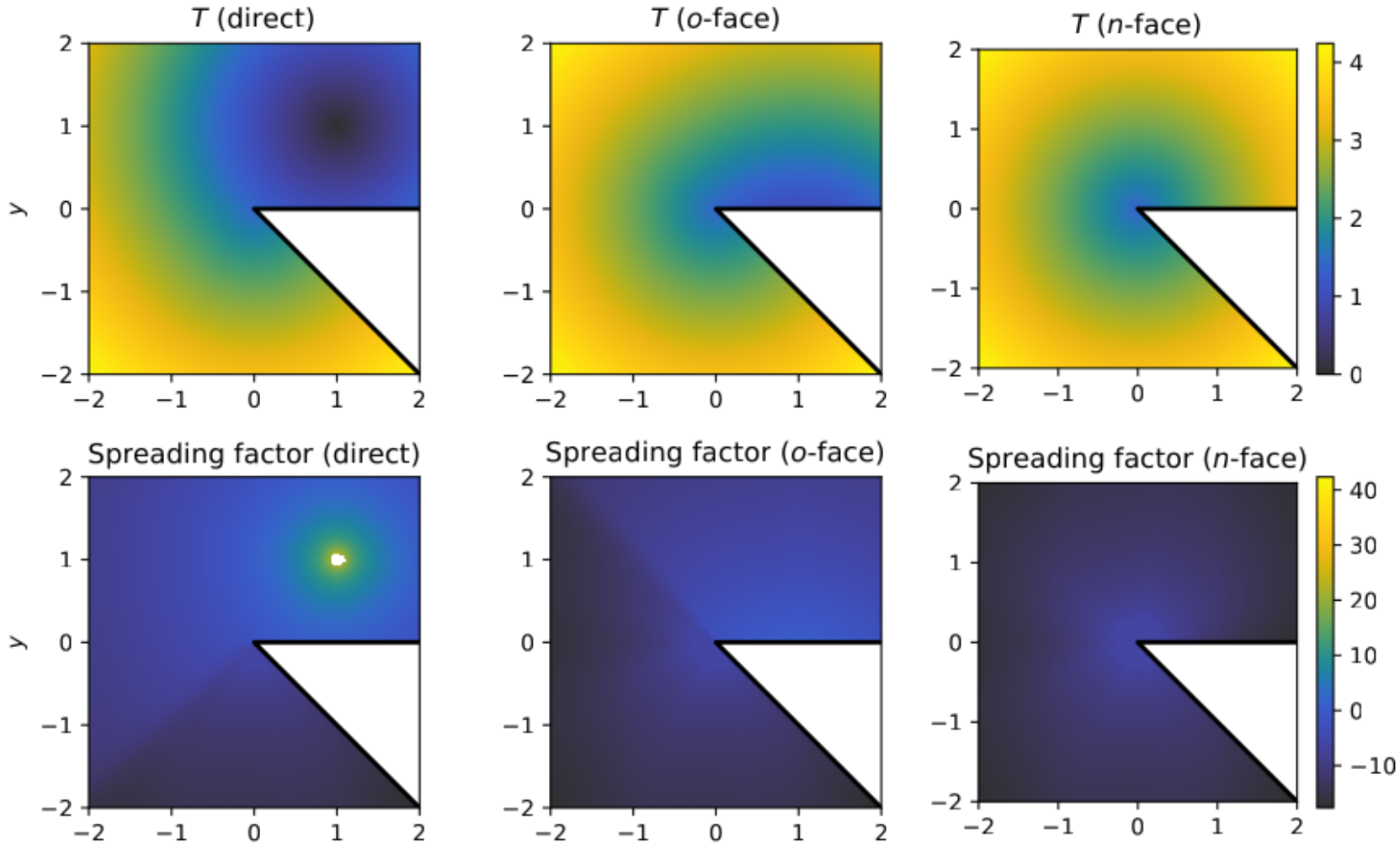


# MA in 3D: numerical shadows using `origin`



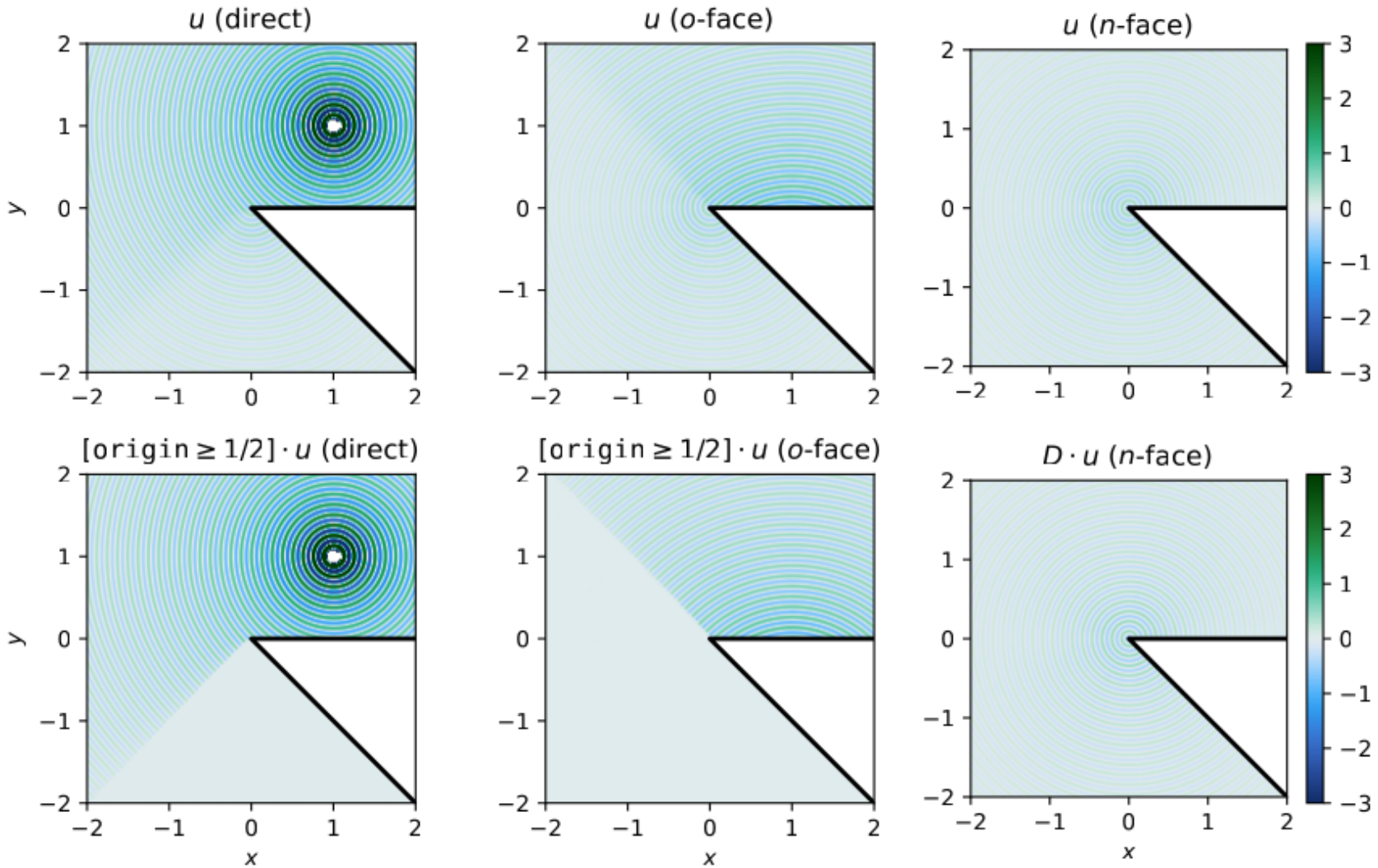
# MA in 3D: total field for wedge

Example of using origin to compute total field for wedge:



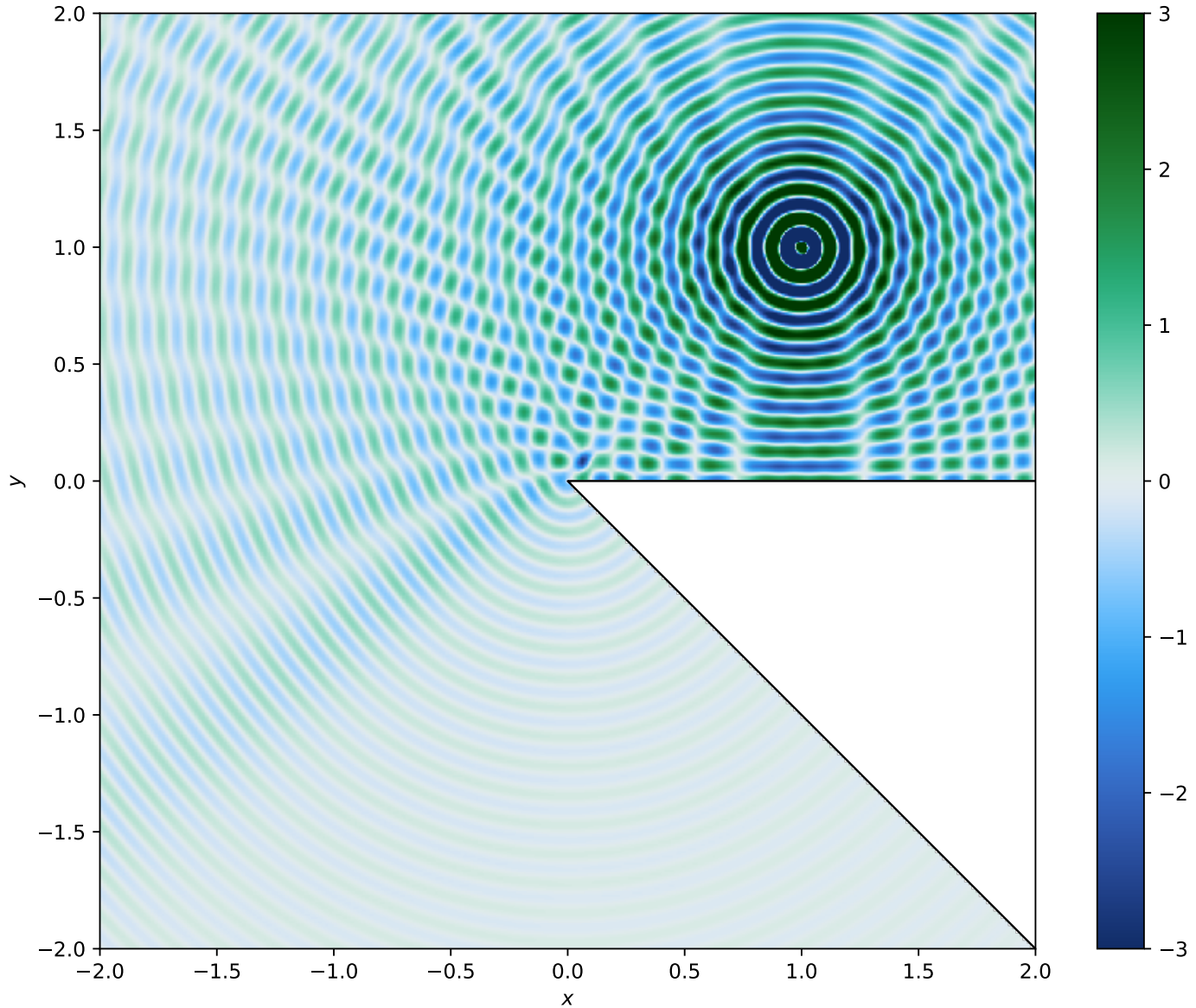
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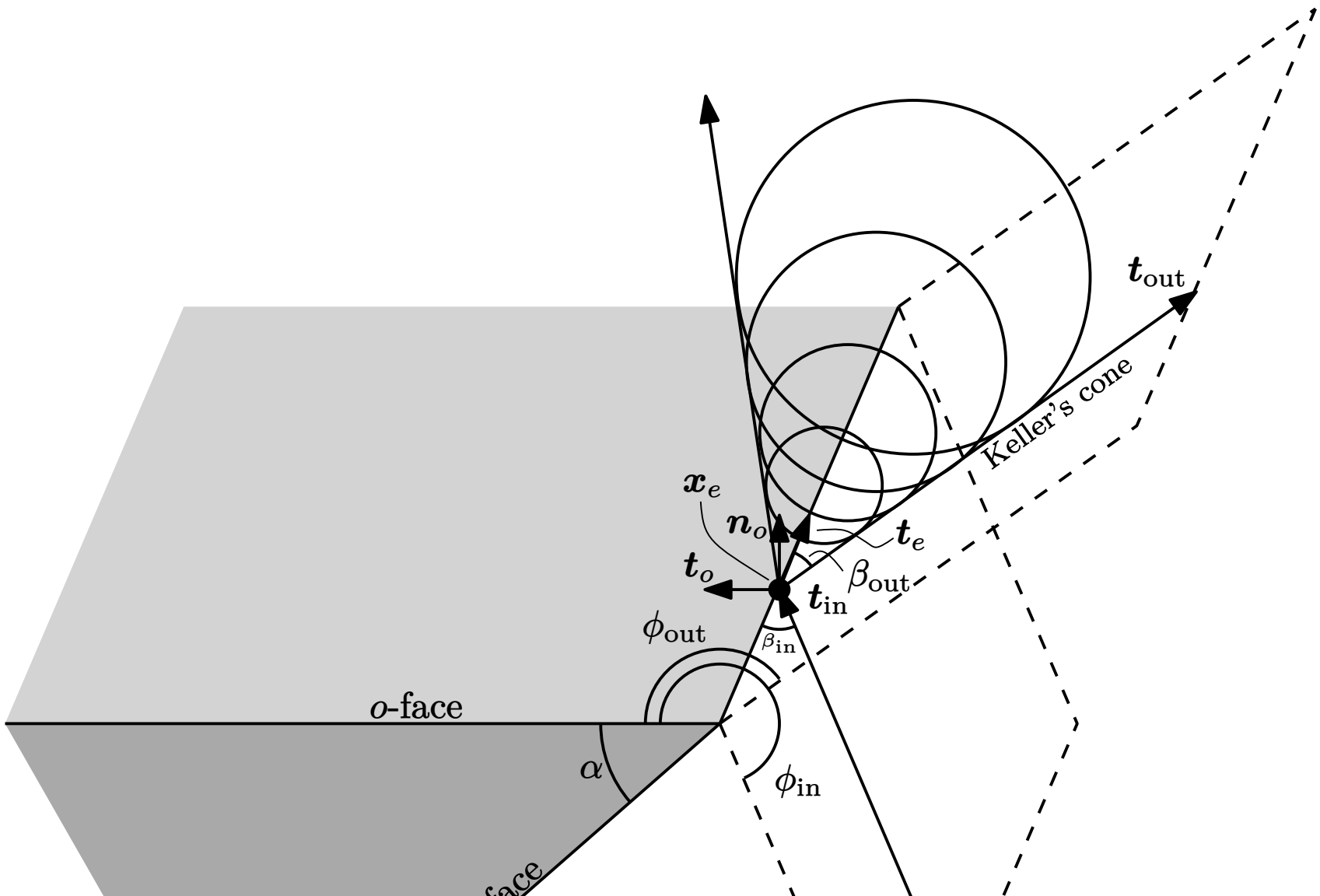
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Example of using origin to compute total field for wedge:

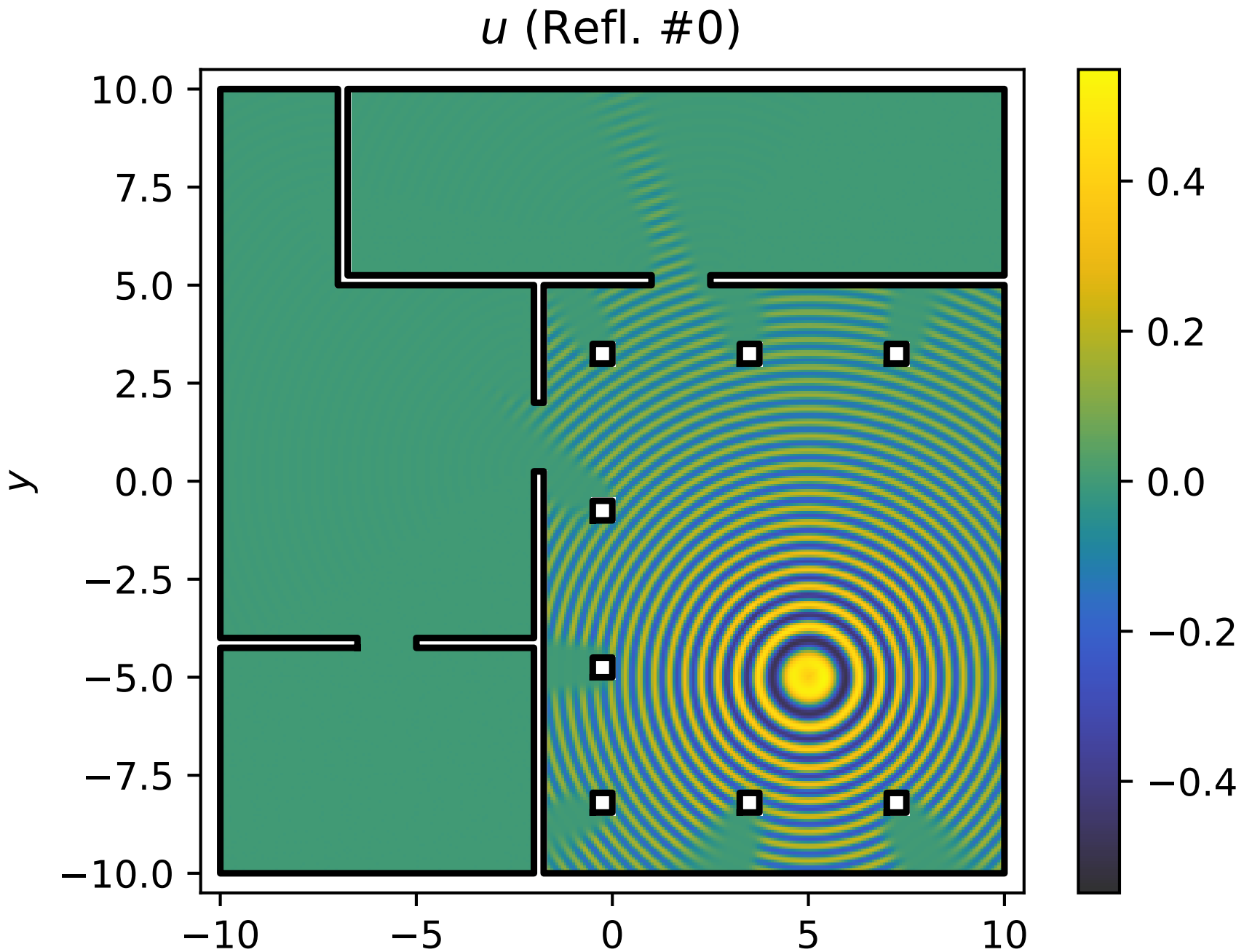


# MA in 3D: uniform theory of diffraction

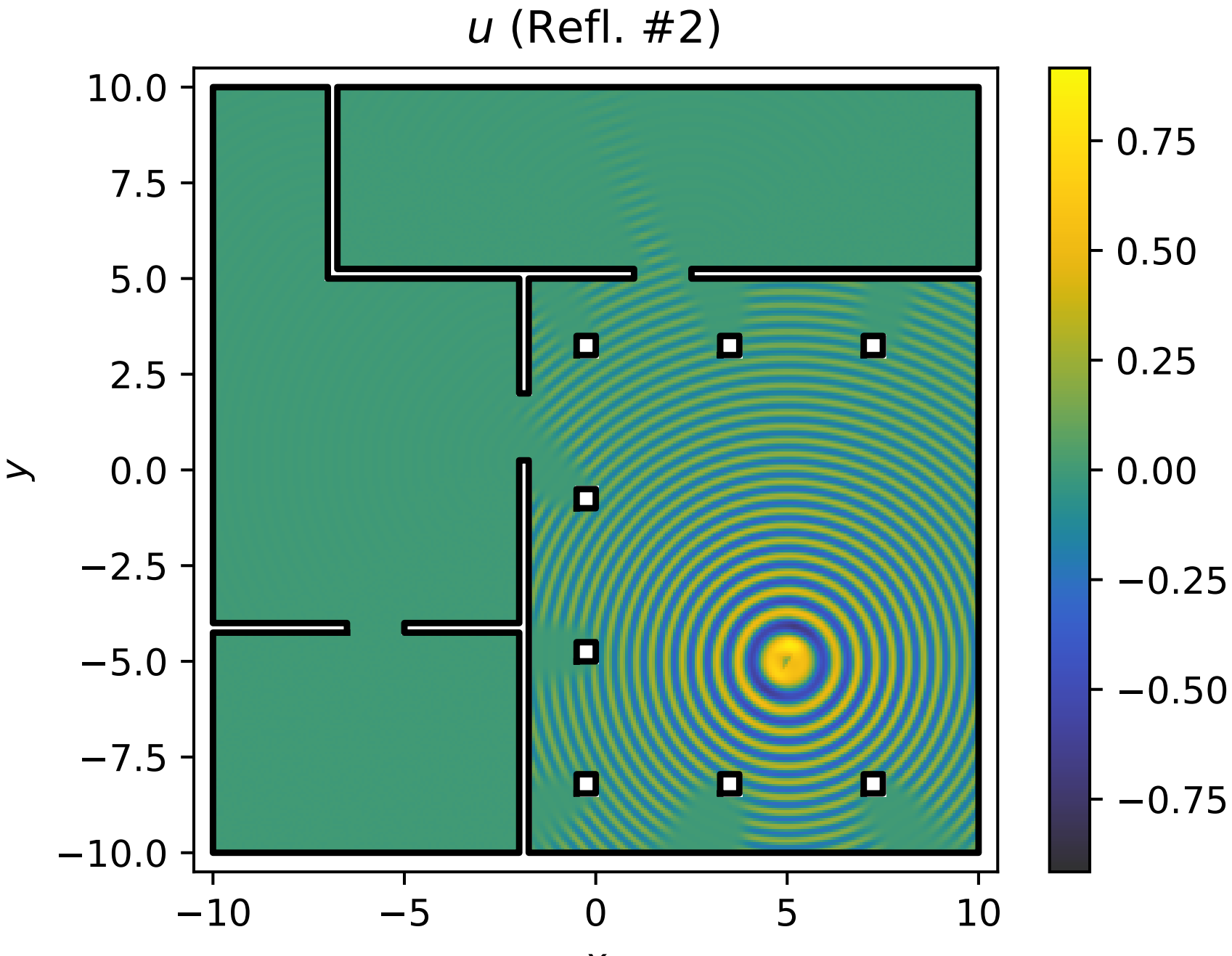
Not exactly correct... need to additionally use the uniform theory of diffraction (UTD) to correct the diffracted amplitude:



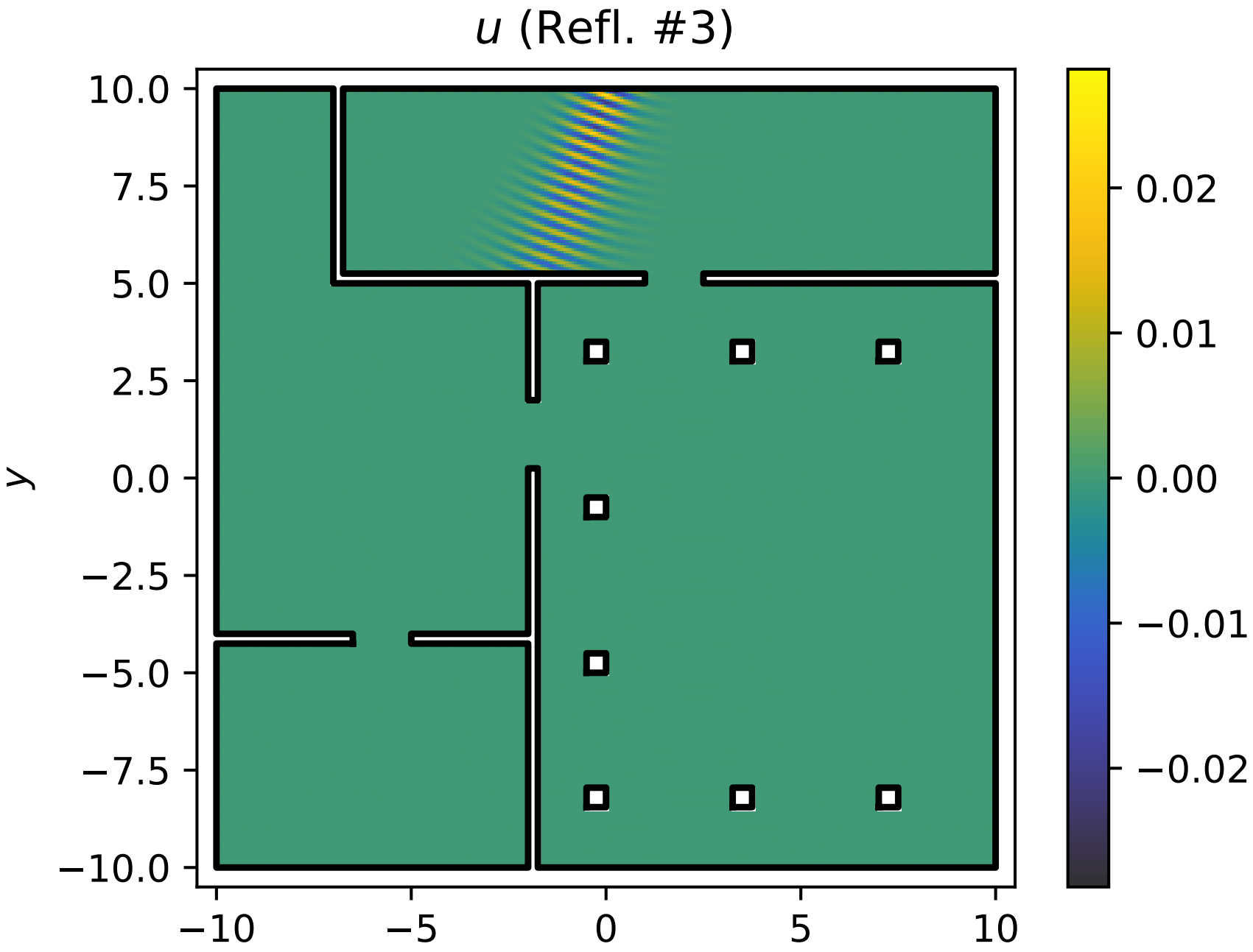
# MA in 3D: room acoustic modeling & RIRs



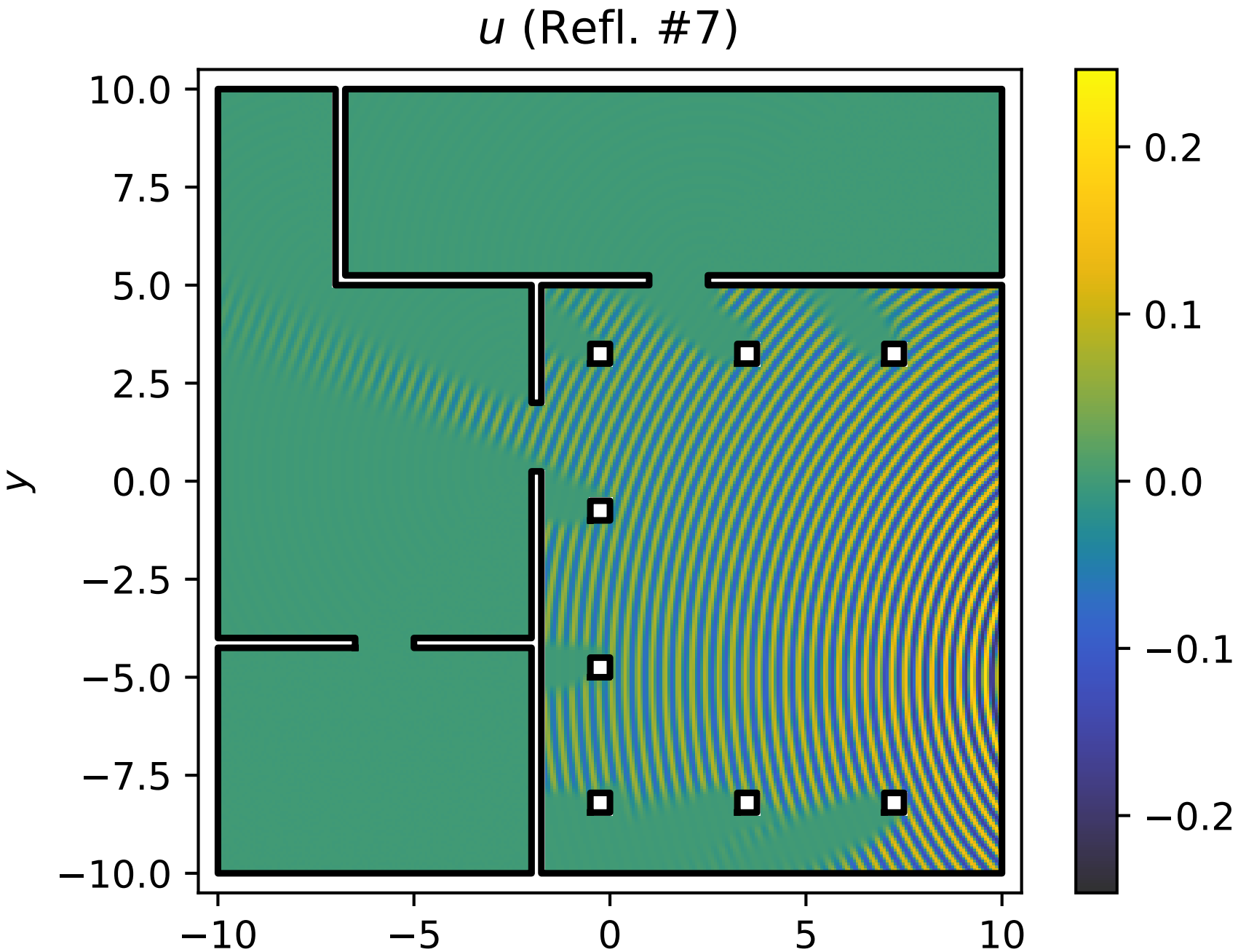
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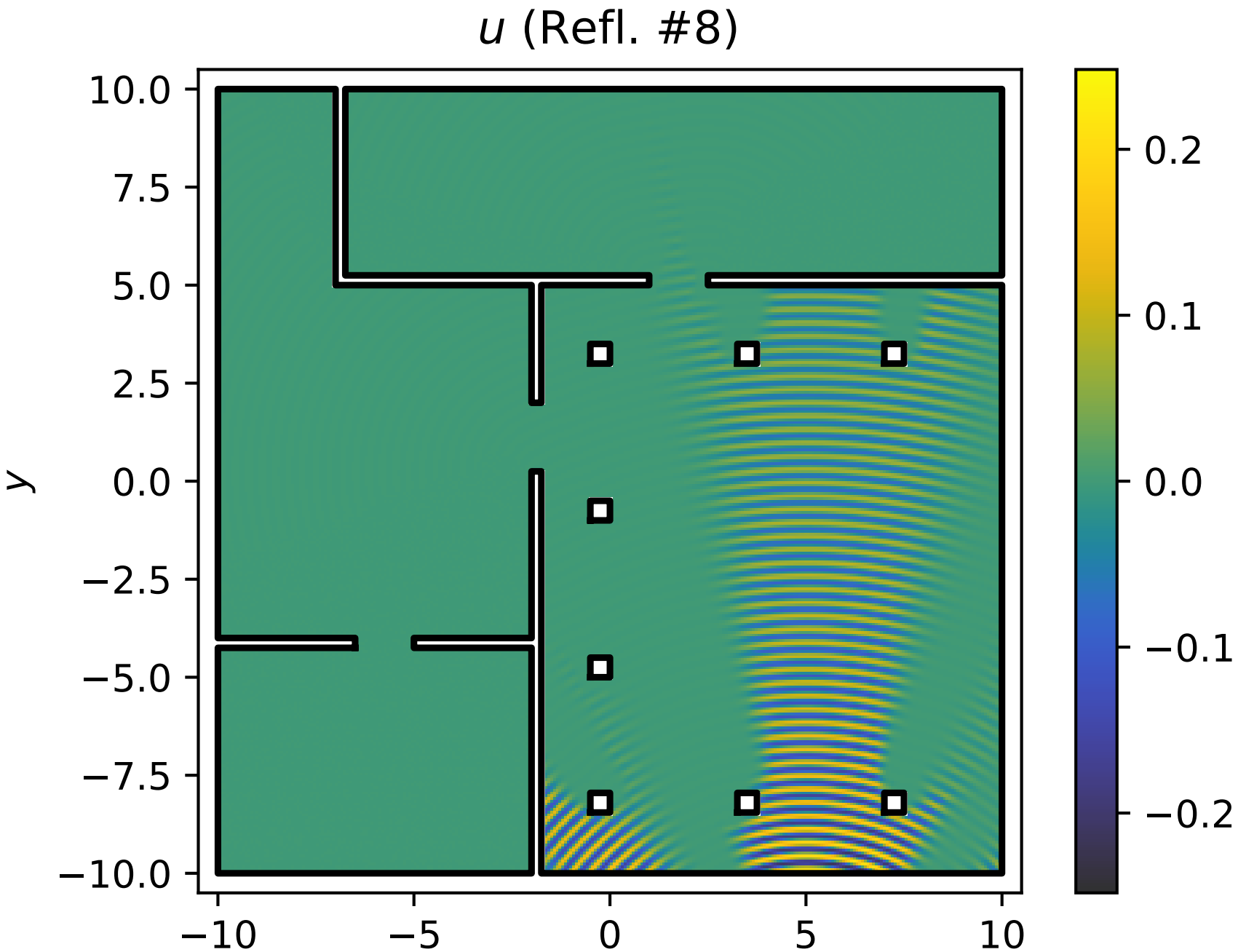
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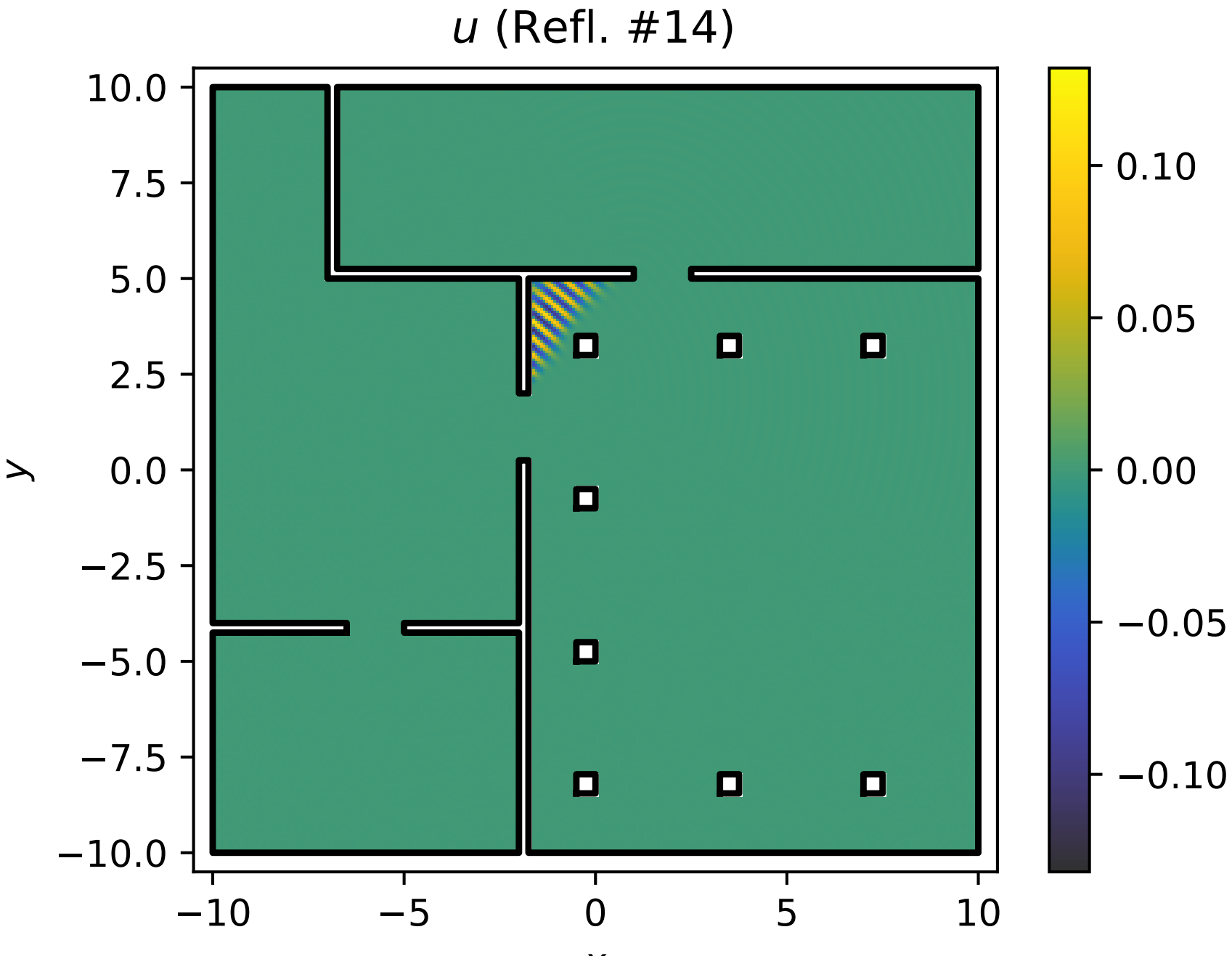
# MA in 3D: room acoustic modeling & RIRs



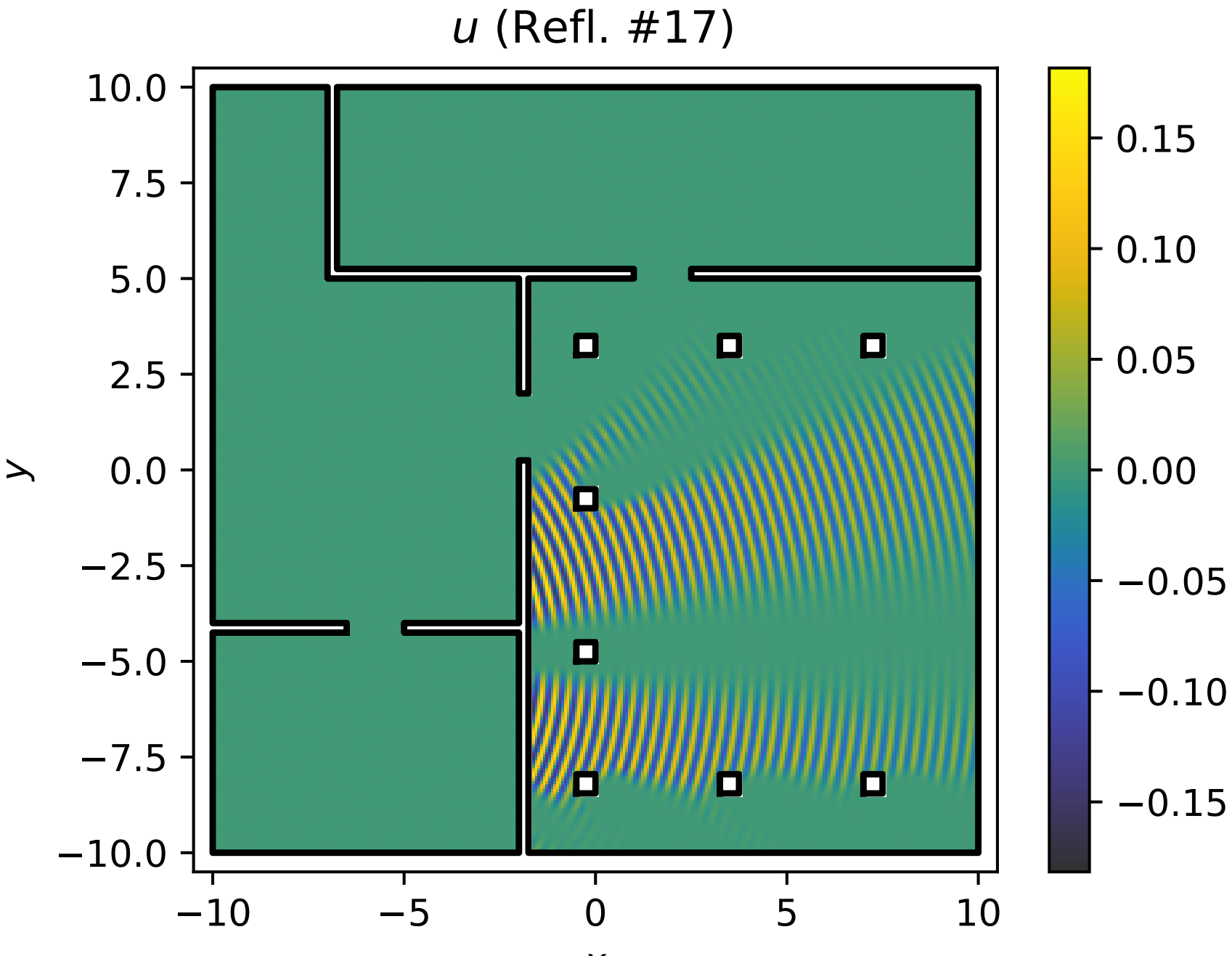
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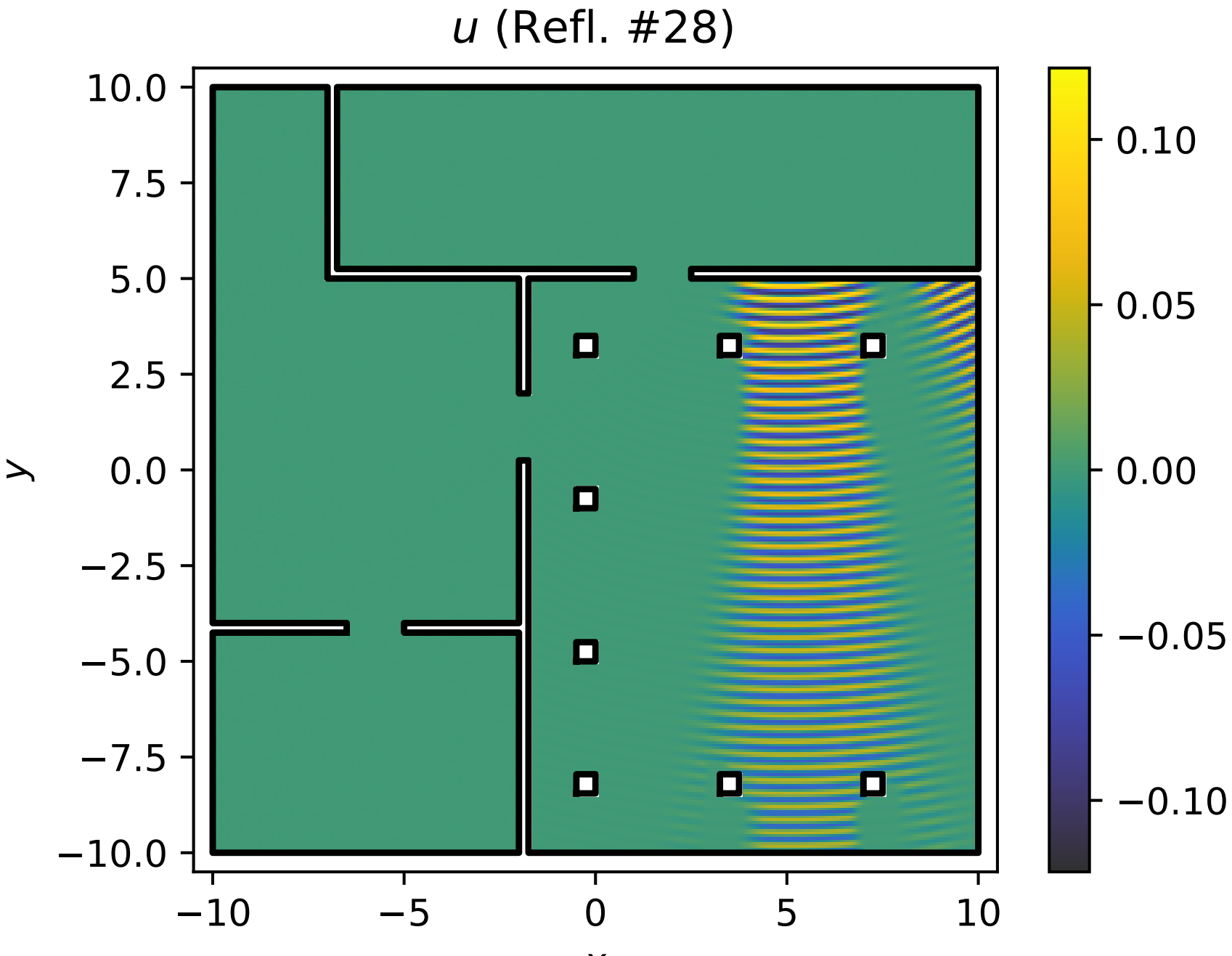
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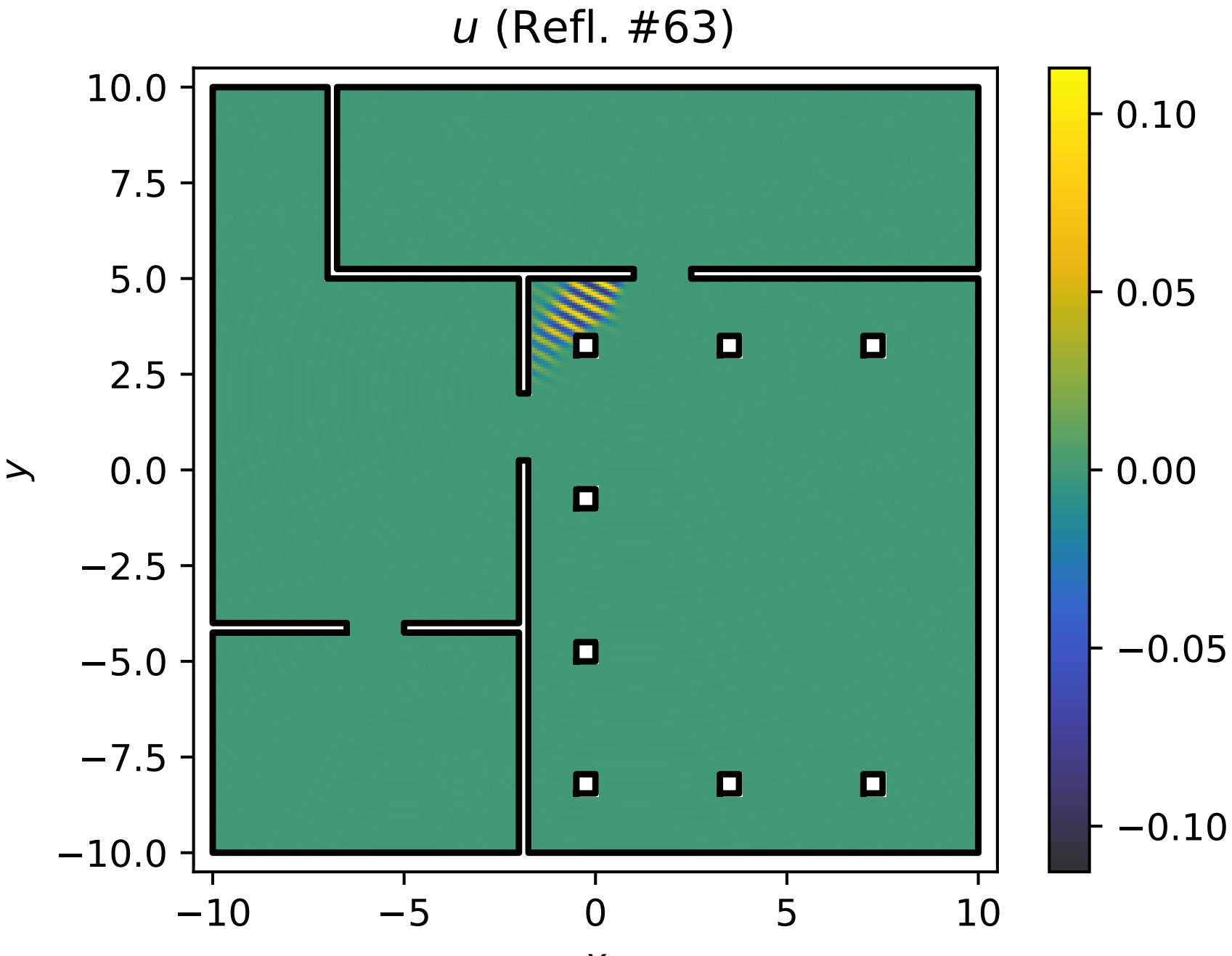
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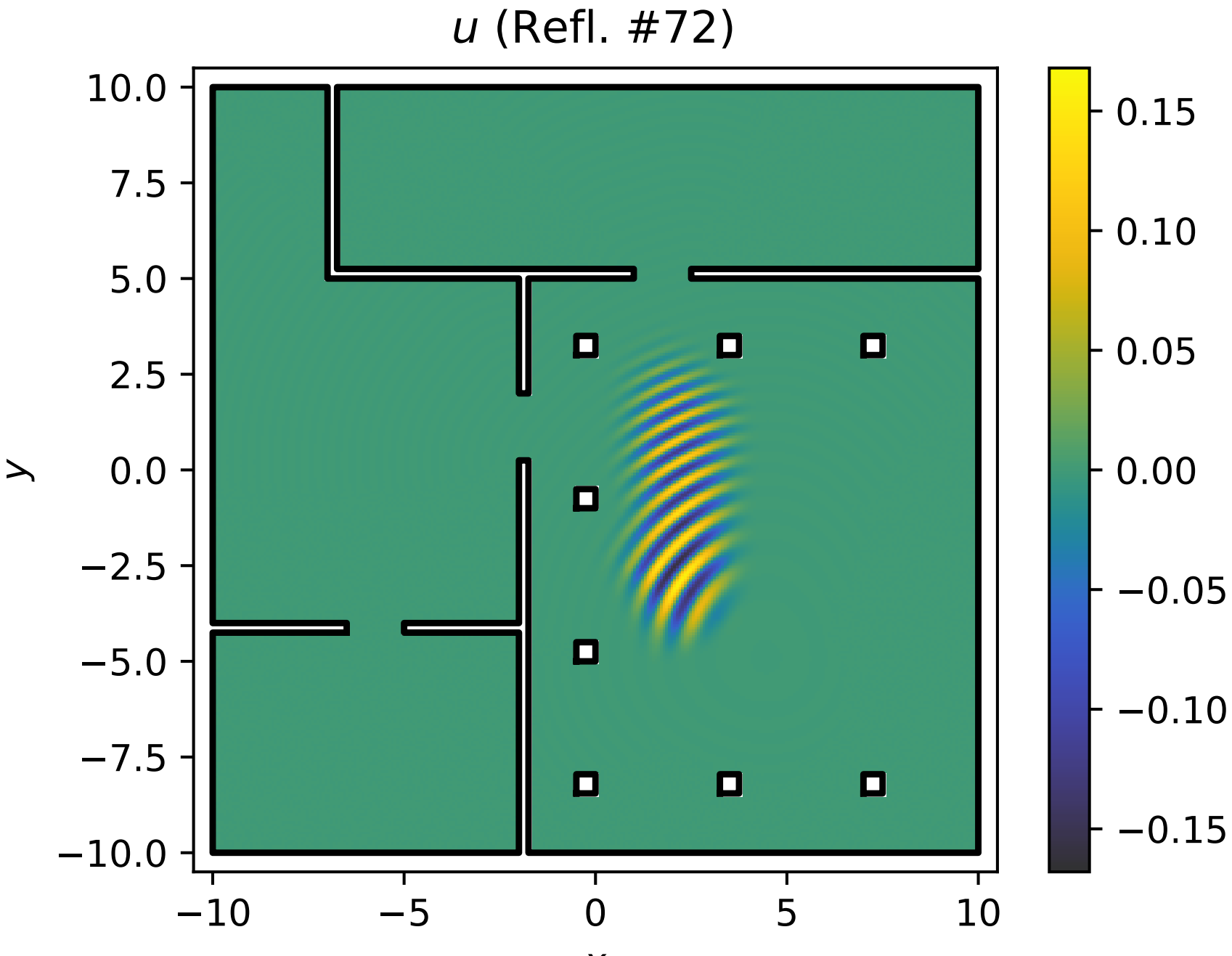
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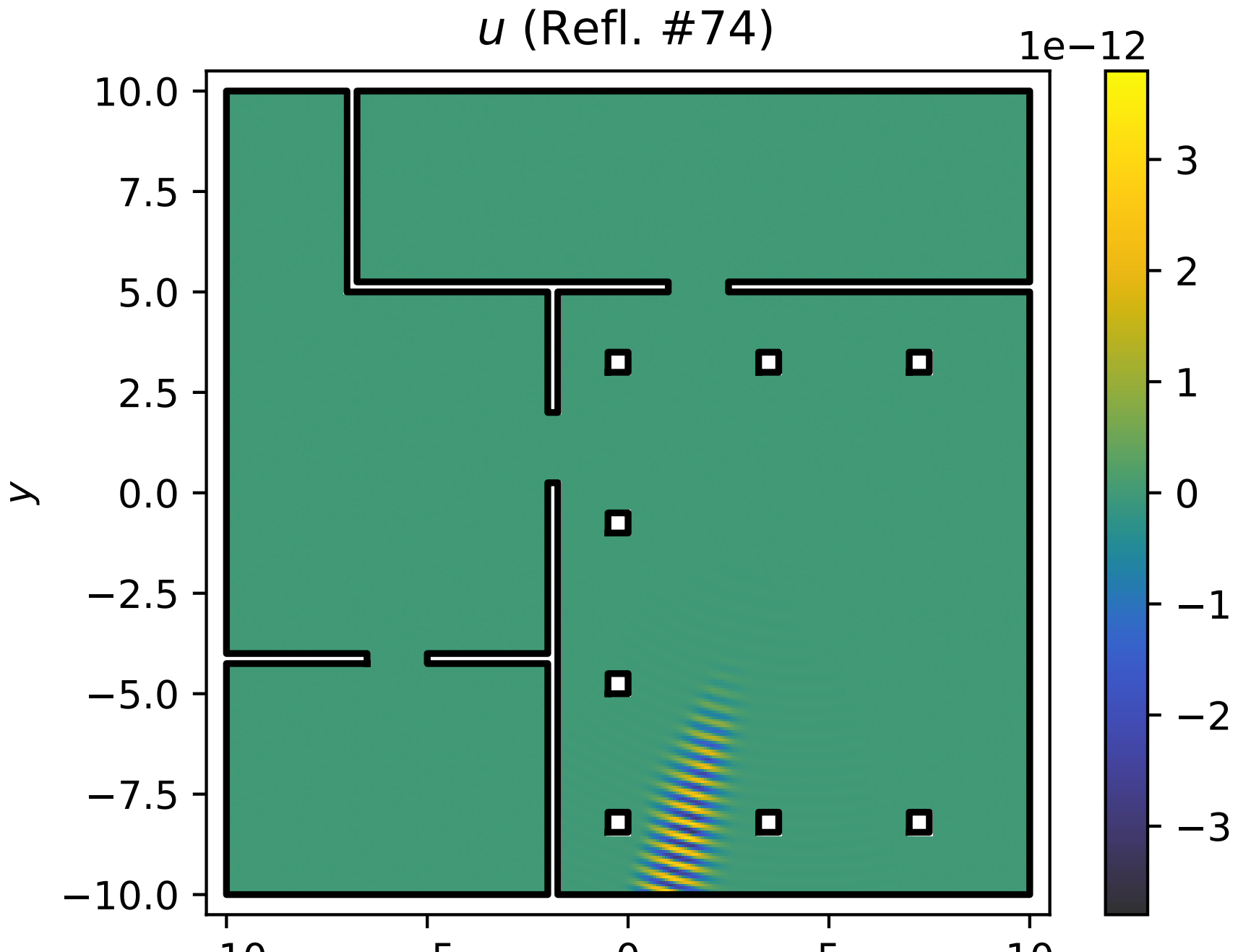
# MA in 3D: room acoustic modeling & RIRs



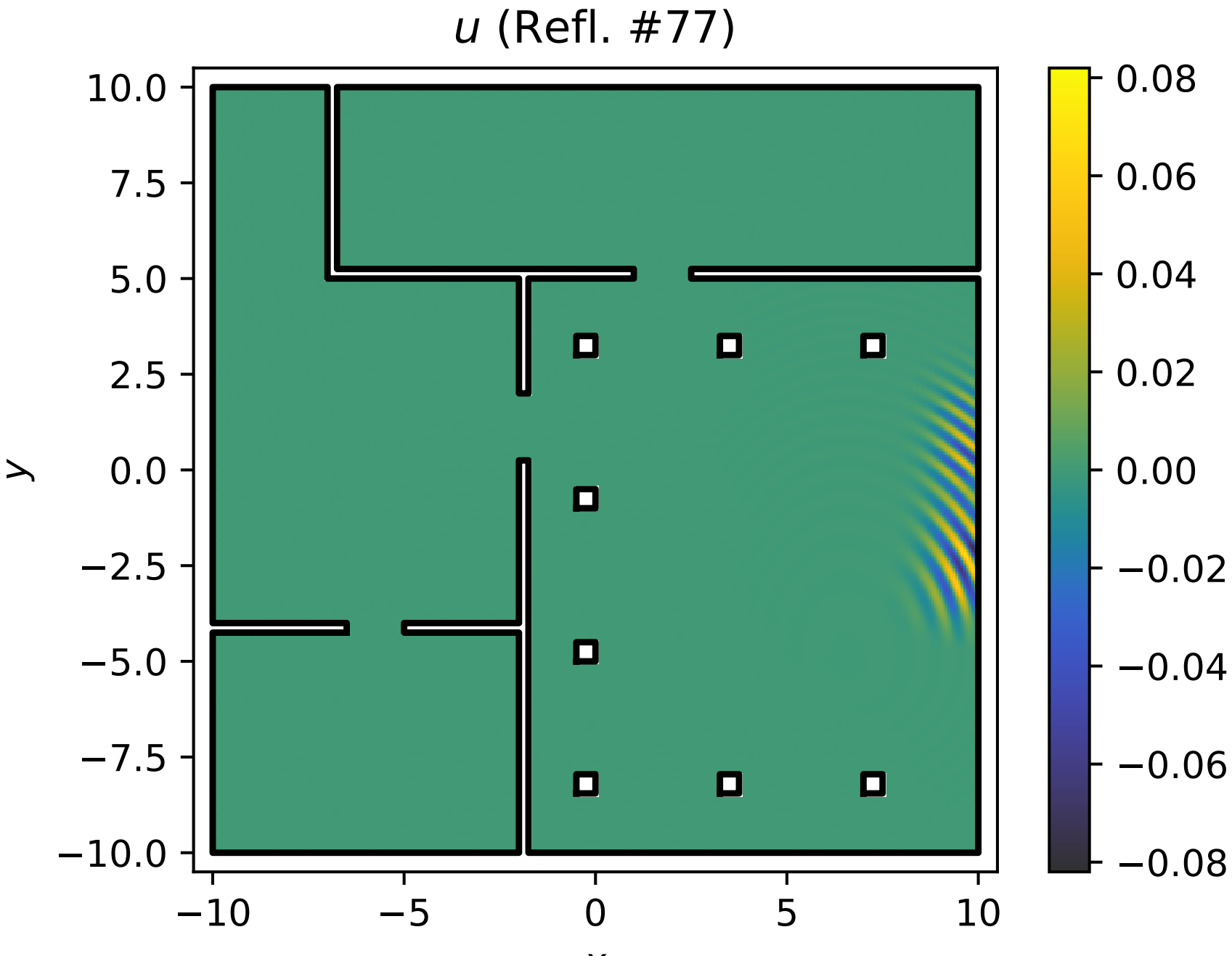
# MA in 3D: room acoustic modeling & RIRs



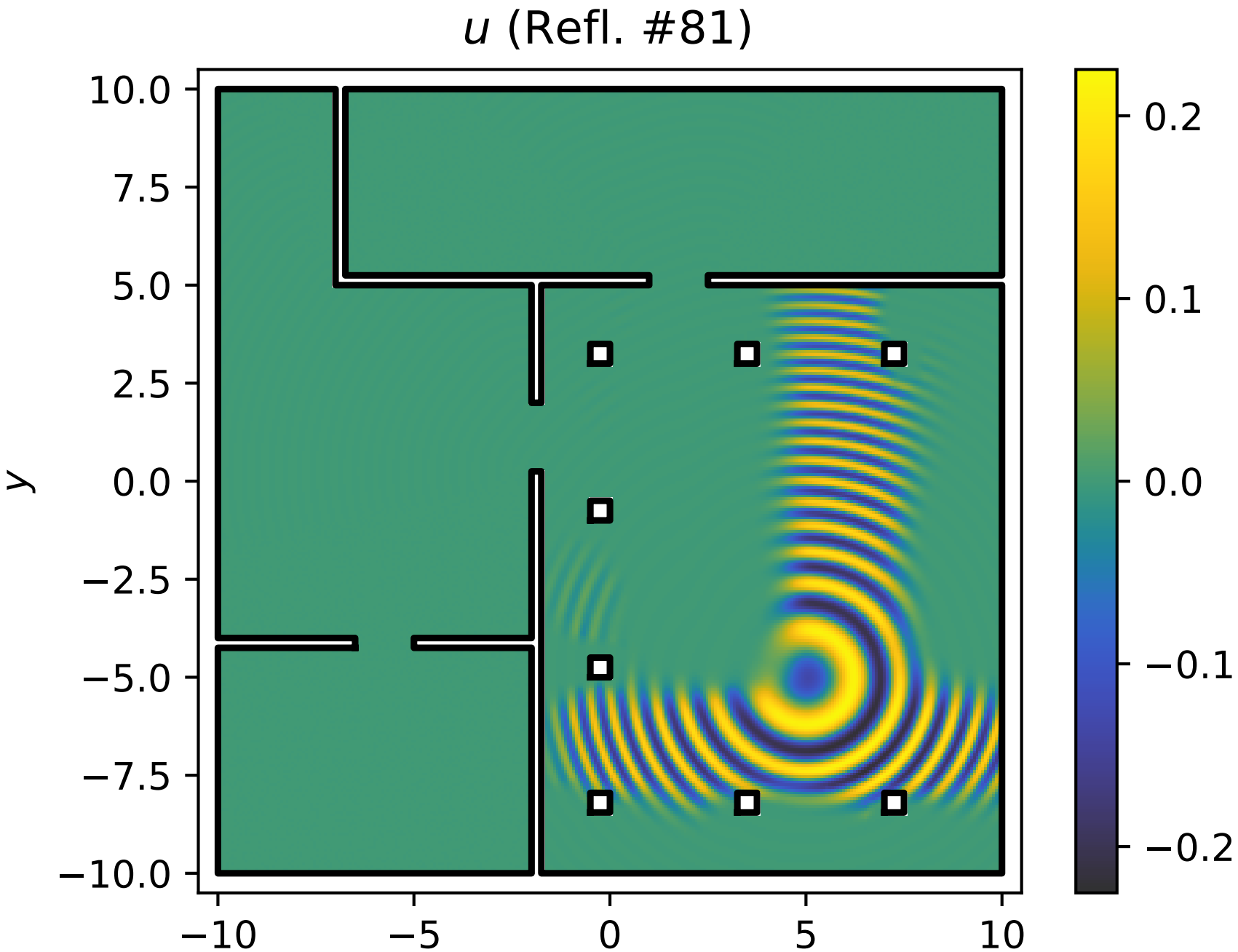
# MA in 3D: room acoustic modeling & RIRs



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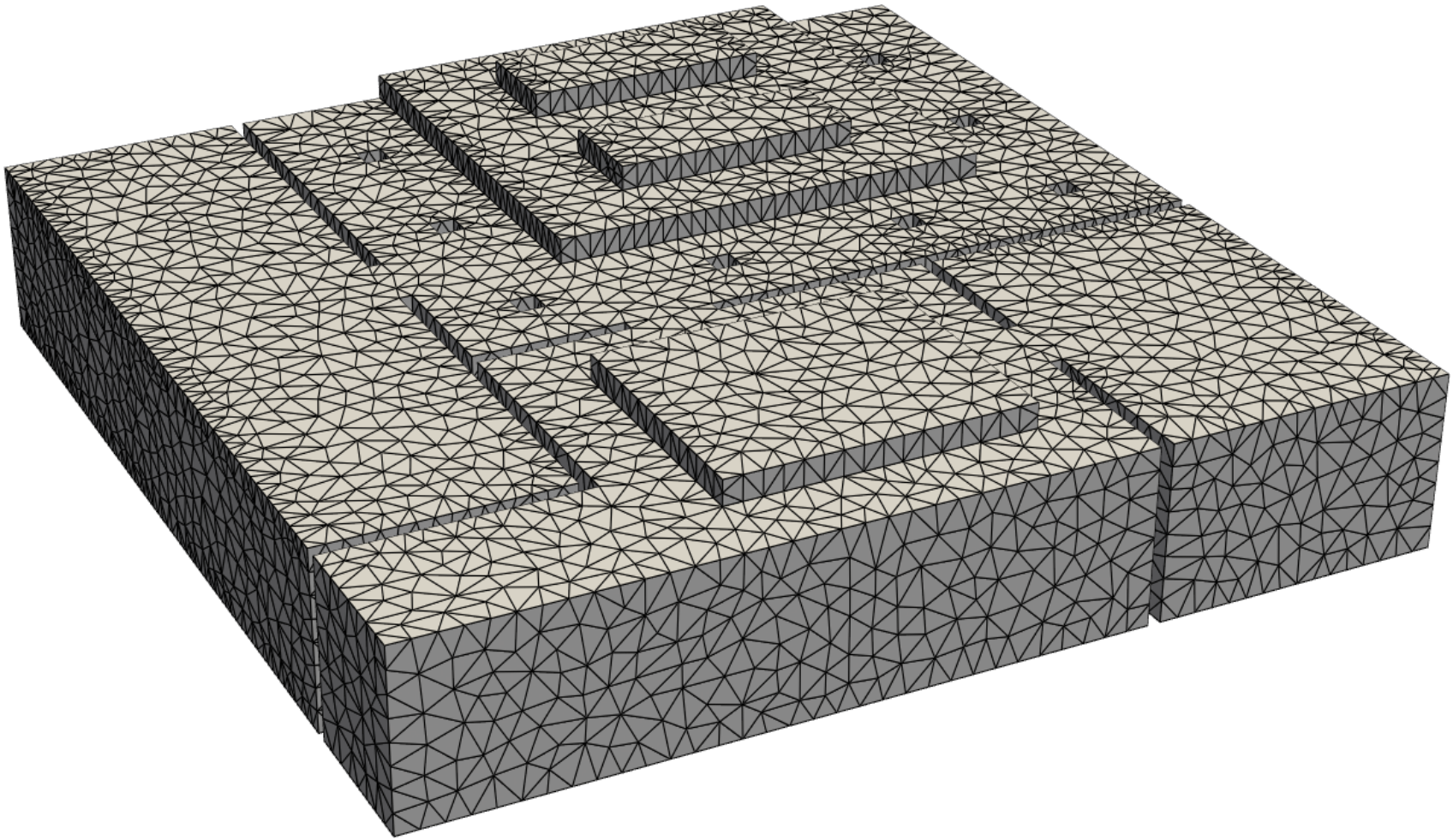


# MA in 3D: room acoustic modeling & RIRs



# Computing room impulse responses (RIRs)

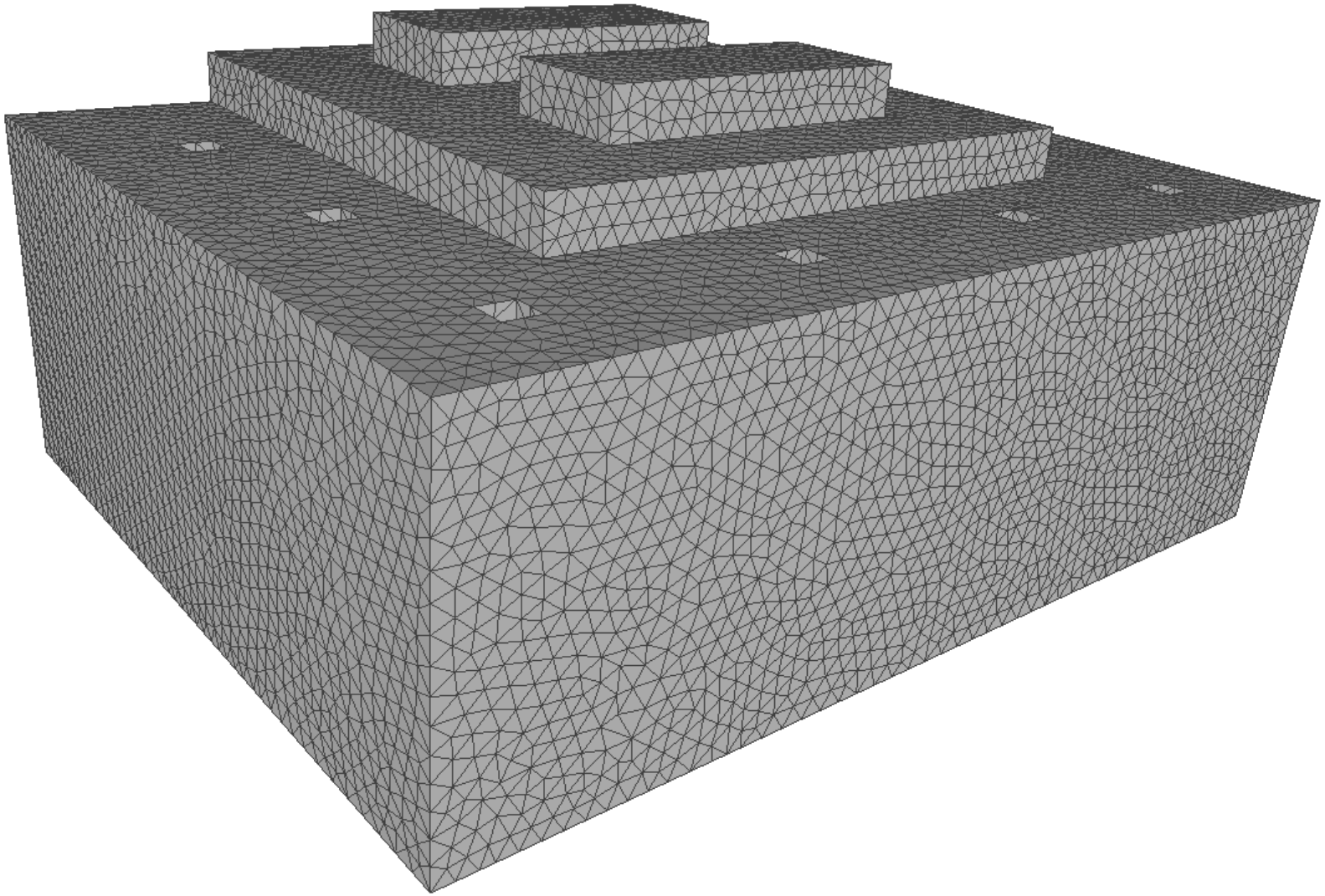
Compare BEM and JMM for computing room impulse response on simple building:



P, Hoover, Zotkin, Duraiswami, 2023.

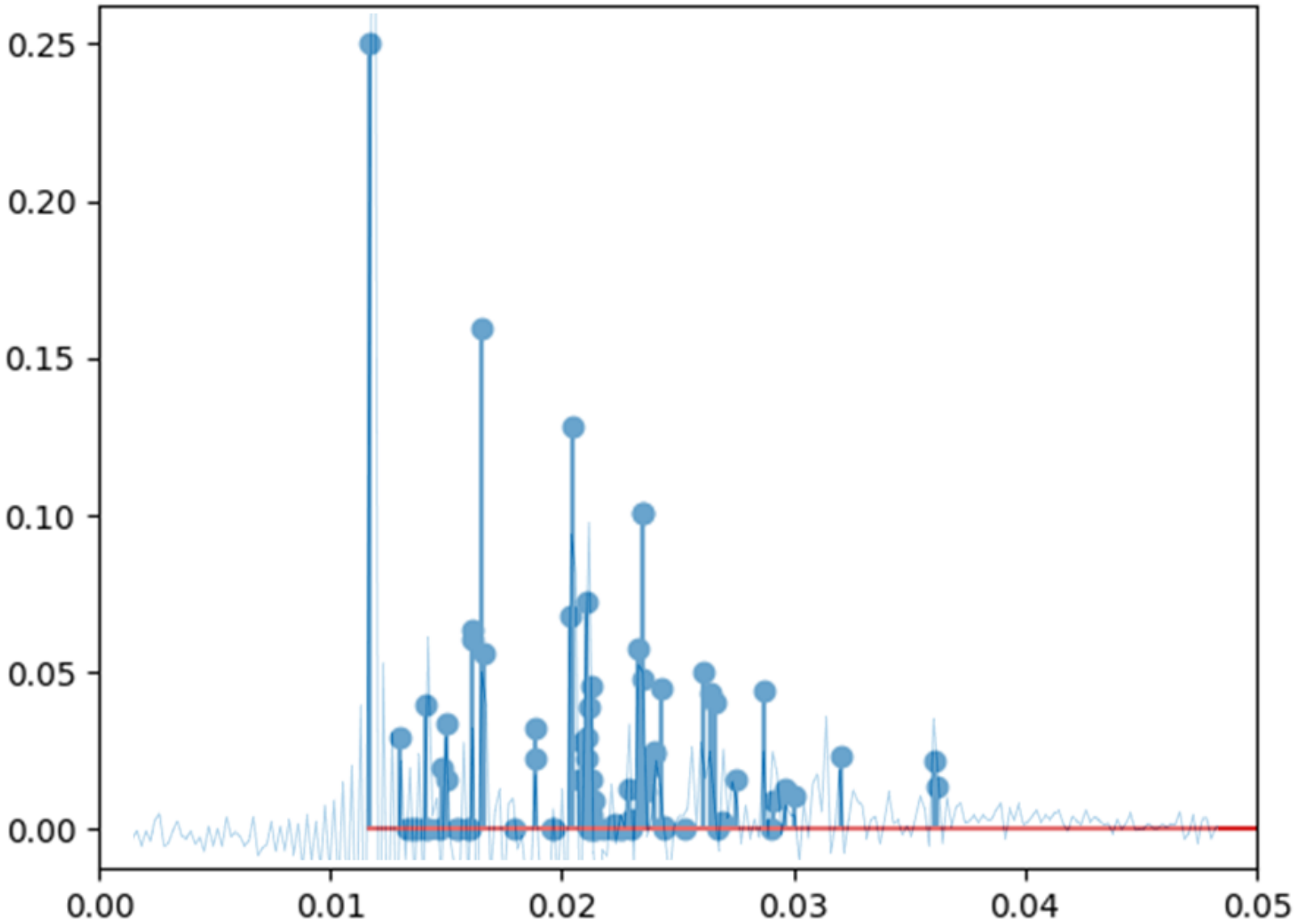
# Computing room impulse responses (RIRs)

Actually, BEM is too slow! ...on one of its rooms:



P. Hoover, Zotkin, Duraiswami, 2023

# Computing room impulse responses (RIRs)




P. Hoover, Zotkin, Duraiswami, 2023

# References

**Samuel F Potter**, Maria K Cameron. *Ordered line integral methods for solving the eikonal equation*, JSC, 2019.

**Samuel F Potter**, Maria K Cameron. *Jet marching methods for solving the eikonal equation*, SISC, 2021.

**Samuel F Potter**, Maria K Cameron, Ramani Duraiswami. *Numerical geometric acoustics: An eikonal-based approach for modeling sound propagation in 3D environments*, JCP, 2023.

**Samuel F Potter**, Monte Hoover, Dmitry Zotkin, Ramani Duraiswami. *Computing acoustic onsets via an eikonal solver*, WASPAA, 2023.